# BANNARI AMMAN INSTITUTE OF TECHNOLOGY

(An Autonomous Institution Affiliated to Anna University, Chennai)
SATHYAMANGALAM - 638 401

**Approved Questions**
S2 Semester

**22GE002 – COMPUTATIONAL PROBLEM SOLVING**

| Q. No. | Questions |
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| | **Unit 1** |
| 1. | A banking system needs to verify whether certain account numbers are prime for security checks. As a software engineer, you are required to write an algorithm that checks whether a given positive integer is a prime number.<br><br>Design a step-by-step algorithm to determine if the number is prime.Ensure your algorithm efficiently handles numbers greater than 1. |
| 2. | A mobile app developer wants to display a list of 12 contact names in alphabetical order in the app's contact list. Currently, the names are unsorted, making it difficult for users to locate a contact quickly.<br><br>Write a step-by-step algorithm using the Bubble Sort technique to sort the contact names alphabetically. Include comparisons, swaps, and iterations until the list is fully sorted. |
| 3. | A fitness app tracks the daily steps of a user for 10 days:<br><br>`[7500, 10200, 8400, 12000, 9800, 15000, 11000, 7000, 9000, 13000]`<br><br>The app needs to:<br><br>1. Calculate the **total steps**.<br>2. Find the **average steps**.<br>3. Count the **number of days the user crossed 10,000 steps**.<br>4. Identify the **maximum steps taken in a day**.<br><br>Write an **algorithm using a `for` loop** to perform all the above operations. |
| 4. | A school library has a small collection of 50 books, each with a unique identification number (Book ID). Occasionally, students request specific books, but the librarian cannot remember the exact location of every book on the shelf. To assist students quickly, the librarian needs a systematic method to **check whether a requested Book ID exists** on the shelf.<br><br>The librarian wants a simple step-by-step procedure that can be followed manually or implemented in a small program to efficiently search for the book.<br><br>Design a simple algorithm that checks whether a given Book ID is present on the shelf and outputs the result. |

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| 5. | A small bakery wants to determine whether a specific item is available in today's menu of 20 items. The system has this pseudocode. Write an algorithm from it.<br><br>**Pseudocode:**<br><br>Input: itemName<br>for each item in menu<br>  if item equals itemName<br>  print "Item Available"<br>  exit<br><br>print "Item Not Available" |
| 6. | A small grocery shop has a list of item prices `[15, 25, 10, 20, 30]`. The cashier wants to check whether an item priced at 20 is available.<br><br>**Algorithm: Check Price Availability**<br><br>1. Read `searchPrice`.<br>2. For each `price` in the list:<br>  ◦ If `price == searchPrice`, print `"Price Found"` and stop.<br>3. If the end of the list is reached without a match, print `"Price Not Found"`.<br><br>Trace the algorithm for `searchPrice = 20` and write the output. |
| 7. | A small online bookstore wants to list all books priced above 500 from a collection `[250, 600, 400, 750, 500]`.<br><br>Write an algorithm to filter these books and display the result. |
| 8. | A college wants to develop a simple system to evaluate student performance. The system should take the marks of a student in three subjects, calculate the total and average, and assign a grade based on the following criteria:<br><br>• Average ≥ 90 → Grade A<br>• Average between 75 and 89 → Grade B<br>• Average between 60 and 74 → Grade C<br>• Average between 50 and 59 → Grade D<br>• Average < 50 → Grade F<br><br>Write the pseudocode to implement this system. |

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| 9. | The pseudocode to check whether a given number is prime or not is given below,<br>BEGIN<br>DECLARE n, i, flag<br>INPUT n<br>IF n <= 1 THEN<br>PRINT "Not Prime"<br>ELSE<br>flag ← 0<br>FOR i ← 2 TO n/2 DO<br>IF n MOD i = 0 THEN<br>flag ← 1<br>BREAK<br>END IF<br>END FOR<br>IF flag = 0 THEN<br>PRINT "Prime Number"<br>ELSE<br>PRINT "Not Prime"<br>END IF<br>END IF<br>END<br><br>Modify the above pseudocode to generate the first n prime numbers. |
| 10. | The algorithm to reverse the digits of a given number is given below,<br><br>**Step 1:** Start the algorithm. **Step 2:** Read the given number **n**.<br>**Step 3:** Initialize reverse to 0.<br>**Step 4:** Check if n > 0.<br>**Step 5:** Find the last digit of the number using n MOD 10.<br>**Step 6:** Multiply reverse by 10.<br>**Step 7:** Add the extracted digit to reverse.<br>**Step 8:** Remove the last digit from n using n DIV 10.<br>**Step 9:** Repeat Steps 4 to 8 until n becomes 0.<br>**Step 10:** Display the value of reverse.<br>**Step 11:** Stop the algorithm.<br><br>Write an equivalent pseudocode to reverse the digits of a given number. |
| 11. | A school system records the marks obtained by a student in an exam. The system needs to evaluate whether the student has passed or failed based on the marks entered. If the marks are 50 or above, the student passes; otherwise, the student fails.<br>Write a pseudocode for the above scenario |

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| 12. | A program is designed to find the sum of digits of a number. However, an incorrect operator is used.<br>BEGIN<br>n ← 123<br>sum ← 0<br>WHILE n > 0 DO<br>digit ← n / 10<br>sum ← sum + digit<br>n ← n / 10<br>END WHILE<br>PRINT sum<br>END<br>Identify the incorrect operator(s) and replace them with the correct ones. |
| 13. | A pseudocode to multiply a number in a loop is given below.<br><br>BEGIN<br>result ← 1<br>FOR i ← 1 TO 5 DO<br>result ← result × 2<br>END FOR<br>PRINT result<br>END<br>Predict the output. |
| 14. | A program is designed to print numbers starting from 1 using a loop. The loop continues as long as the value of the variable is less than or equal to 5.<br>BEGIN<br>i ← 1<br>WHILE i ≤ 5 DO<br>PRINT i<br>i ← i + 1<br>END WHILE<br>END<br>Identify the loop termination condition. When will the loop stop executing? |
| 15. | A health app tracks daily physical activity. It asks the user to input the number of steps taken in a day and the time (in minutes) spent walking. The app categorizes the user's activity level as follows:<br>• Steps less than 1 → Invalid Input<br>• 1 to 4999 steps → Activity Level = 0<br>• 5000 to 9999 steps → Activity Level = 25<br>• 10000 to 14999 steps → Activity Level = 50<br>• 15000 steps and above → Activity Level = 100<br><br>Using the above scenario, **generate pseudocode** that takes the user's input, validates it, and outputs the appropriate activity level. |

16.

A delivery company tracks daily parcel deliveries for a week (7 days). For each day, the system asks the user to input the **number of parcels delivered** and **time (in minutes) spent delivering**. The system calculates:

- **Total parcels delivered in the week**
- **Average delivery time per valid day**
- **Highest single-day parcel delivery**

It also categorizes weekly performance based on total parcels delivered:

- Less than 200 → Low
- 200 to 499 → Medium
- 500 to 799 → High
- 800 and above → Very High

If the user enters **negative parcels or time**, the program should print `"Invalid Input"` and skip that day in calculations.

Write an **algorithm** to implement the above weekly tracking and calculations.

17.

An online examination system manages student login, verification, monitoring, and result evaluation. The system performs the following steps:

- The user logs in and undergoes **identity verification**.
- If verification fails, the exam session is terminated.
- If successful, the exam starts and the system monitors for **suspicious activity**.
- If suspicious activity is detected, the system checks its severity:
    - Low severity → Warning is given, and the exam continues.
    - High severity → Exam is paused and the administrator is notified.
- If no suspicious activity occurs, the student continues the exam.
- When time ends, the exam is automatically submitted and evaluated.
- If marks are above the passing criteria → Display **“Pass”**
- Otherwise → Display **“Fail”**

Draw a **flowchart** for the above scenario.

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| 18. | A warehouse system tracks items packed over 5 days using the following pseudocode:<br><br>START<br>totalItems ← 0<br>maxItems ← 0<br><br>FOR day FROM 1 TO 5<br>  Input items<br><br>  IF items < 0 THEN<br>  items ← 0<br>  ENDIF<br><br>  totalItems ← totalItems + items<br><br>  IF items > maxItems THEN<br>  maxItems ← items<br>  ENDIF<br>ENDFOR<br><br>average ← totalItems / 5<br><br>Print "Total:", totalItems<br>Print "Average:", average<br>Print "Max:", maxItems<br><br>END<br><br>Modify the above pseudocode to satisfy the following new conditions:<br><br>1. If `items > 500`, treat it as **invalid** and skip that day (do not include in total or average).<br>2. Calculate **average only for valid days** (not fixed 5 days).<br>3. If no valid input is given, print `"No valid data"` instead of average. |
| 19. | A cafeteria system records meal orders for 4 customers. For each customer, the system takes the number of meals ordered and applies the following logic:<br><br>• If meals ≤ 0 → Ignore the entry<br>• If meals between 1 and 5 → Add to total orders<br>• If meals more than 5 → Add only 5 meals to total (limit applied)<br><br>At the end, the system prints the **total number of valid meals ordered**.<br><br>Given the inputs:<br>**3, 7, -2, 5**<br><br>Find the final total meals ordered and identify why meal limits are applied in the system. |

20. A system processes item quantities using the following pseudocode:

```
START

total ← 0

FOR i FROM 1 TO 4
  Input qty

  IF qty <= 0 THEN
     CONTINUE
  ELSE IF qty > 5 THEN
     total ← total + 5
  ELSE
     total ← total + qty
  ENDIF
ENDFOR
Print total
END
```

Given the inputs:
**3, 7, -2, 5**

Find the **final output** printed by the program.

21.

A library system processes book return days using the following pseudocode:

```
START

fine ← 0

FOR i FROM 1 TO 3

   Input daysLate

   IF daysLate <= 0 THEN

      fine ← fine + 0

   ELSE IF daysLate <= 5 THEN

      fine ← fine + (daysLate * 2)

   ELSE

      fine ← fine + 10

   ENDIF

ENDFOR

Print fine

END
```

Explain the following concepts used in the above pseudocode:

1. The use of conditional statements (IF–ELSE IF–ELSE)
2. The reason for applying a fixed fine when daysLate > 5

22. Scenario decomposition is a fundamental technique in divide-and-conquer algorithms, where a problem is broken into smaller, manageable sub-problems. The Merge Sort algorithm sorts an array by recursively dividing it into two halves until each subarray contains only one element and then merging them back in sorted order.

Given an array:

**A = [15, 3, 9, 7, 1, 12, 6, 4]**

Perform a **full decomposition** of the given array using Merge Sort. Recursively split the array until each subarray contains only one element and illustrate the complete recursive structure at each stage.

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| 23. | Consider the following pseudocode:<br><br>START<br>sum ← 0<br>count ← 0<br>FOR i FROM 1 TO 5<br>  Input marks<br><br>  IF marks < 0 OR marks > 100 THEN<br>  Print "Invalid Marks"<br>  ELSE<br>  sum ← sum + marks<br>  count ← count + 1<br>  ENDIF<br>ENDFOR<br>IF count > 0 THEN<br>  average ← sum / count<br>ELSE<br>  average ← 0<br>ENDIF<br>IF average >= 75 THEN<br>  grade ← "Distinction"<br>ELSE IF average >= 50 THEN<br>  grade ← "Pass"<br>ELSE<br>  grade ← "Fail"<br>ENDIF<br>Print average<br>Print grade<br>END<br><br>Decompose the above program into smaller components and:<br><br>1. Identify the **inputs, outputs, and variables** used.<br><br>2. Break the program into **logical modules/steps**.<br><br>3. Identify the **key decision-making conditions**. |
| 24. | A retail chain observes frequent stock-outs in their stores despite regular inventory audits. Store managers report delayed deliveries, inaccurate stock records, and lack of demand forecasting.<br><br>Break down the stock-out issue using scenario decomposition, identify two sub-problems, and explain how solving them individually improves efficiency. |
| 25. | A software development team struggles to deliver a mobile application on time. Issues include unclear client requirements, team miscommunication, and inefficient task allocation.<br><br>Decompose the project delays into sub-problems, select one sub-problem as priority, and suggest a method to resolve it. |

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| 26. | A food delivery company notices that orders are frequently delivered late. Customers complain about delayed deliveries, while delivery personnel report unclear route assignments and long waiting times at restaurants.<br><br>Using scenario decomposition, identify **the main root cause of delays** and **one sub-problem that should be addressed first**. |
| 27. | A school is organizing its annual sports day with 500 students participating. Teachers face challenges in scheduling events, assigning referees, allocating equipment, and arranging different sports venues. Students and parents often get confused about timings, locations, and participation criteria.<br><br>Decompose this problem into **three sub-problems** that can be addressed individually. |
| 28. | A system processes 5 daily expenses. The system must follow these constraints:<br><br>• Only positive values should be considered<br>• Maximum allowed expense per entry is ₹10,000<br>• Total expense should not exceed ₹30,000<br>• If the limit exceeds, stop processing further inputs<br>• Finally, display total expense and number of valid entries<br><br>1. Identify **any two constraints** that affect input processing. (2 Marks)<br>2. Identify **any two conditions that control program flow**. (2 Marks)<br>3. State **when the system will terminate early**. (1 Mark) |
| 29. | A college system processes the marks obtained by a student in multiple subjects. It calculates the total and average marks and then assigns a grade based on predefined conditions. The system performs multiple steps to evaluate the student's performance.<br><br>Identify any two subtasks involved in this scenario. |
| 30. | A school system evaluates a student's performance based on the marks obtained. It uses conditions to assign grades such as A, B, or C depending on the average marks. The system makes decisions using IF statements to determine the appropriate grade.<br><br>Predict the type of control flow used and specify its type. |
| 31. | A project team is consistently failing to meet deadlines for assigned tasks. This is affecting overall productivity and delaying project delivery. The management wants to identify the root causes and improve team performance.<br><br>Break the problem into any two sub-problems and suggest one solution for each. |

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| 32. | A small factory is taking too much time to produce goods, and orders are getting delayed. This is causing customer dissatisfaction and reduced profits. The owner wants to find the problem and make the production process faster.<br><br>Identify the undesirable state and the corresponding goal. Also, mention one possible solution approach. |
| 33. | A program is required to sort a list of 10 numbers in ascending order. The developer finds it difficult to directly write the solution due to multiple steps involved in sorting. To solve this efficiently, the problem needs to be broken down into smaller manageable steps.<br><br>Apply the steps of decomposition to the above scenario and explain how each step can be used to sort the numbers. |
| 34. | A web application is experiencing slow performance and frequent crashes when multiple users access it simultaneously. The development team is unsure whether the issue is due to inefficient code, server overload, database queries, or network latency. They want to clearly analyze the problem before optimizing the system.<br><br>Analyze the given scenario using appropriate problem-understanding strategies. Also, explain how rephrasing the problem, using visual representations, identifying knowns and unknowns, and asking “Why” and “How” questions help in understanding the issue. |
| 35. | A digital library system stores book IDs in a sorted list for fast searching. When a user searches for a specific book ID, the system does not check each ID one by one. Instead, it repeatedly checks the middle element of the list and decides whether to search in the left half or right half. This process continues until the required book ID is found or the search space becomes empty.<br><br>Given sorted book IDs: [101, 115, 128, 134, 146, 159, 172, 185, 193, 207]<br><br>Apply decomposition to solve the problem. Break the process into steps and find the position of the book ID 172 by showing the steps clearly. |

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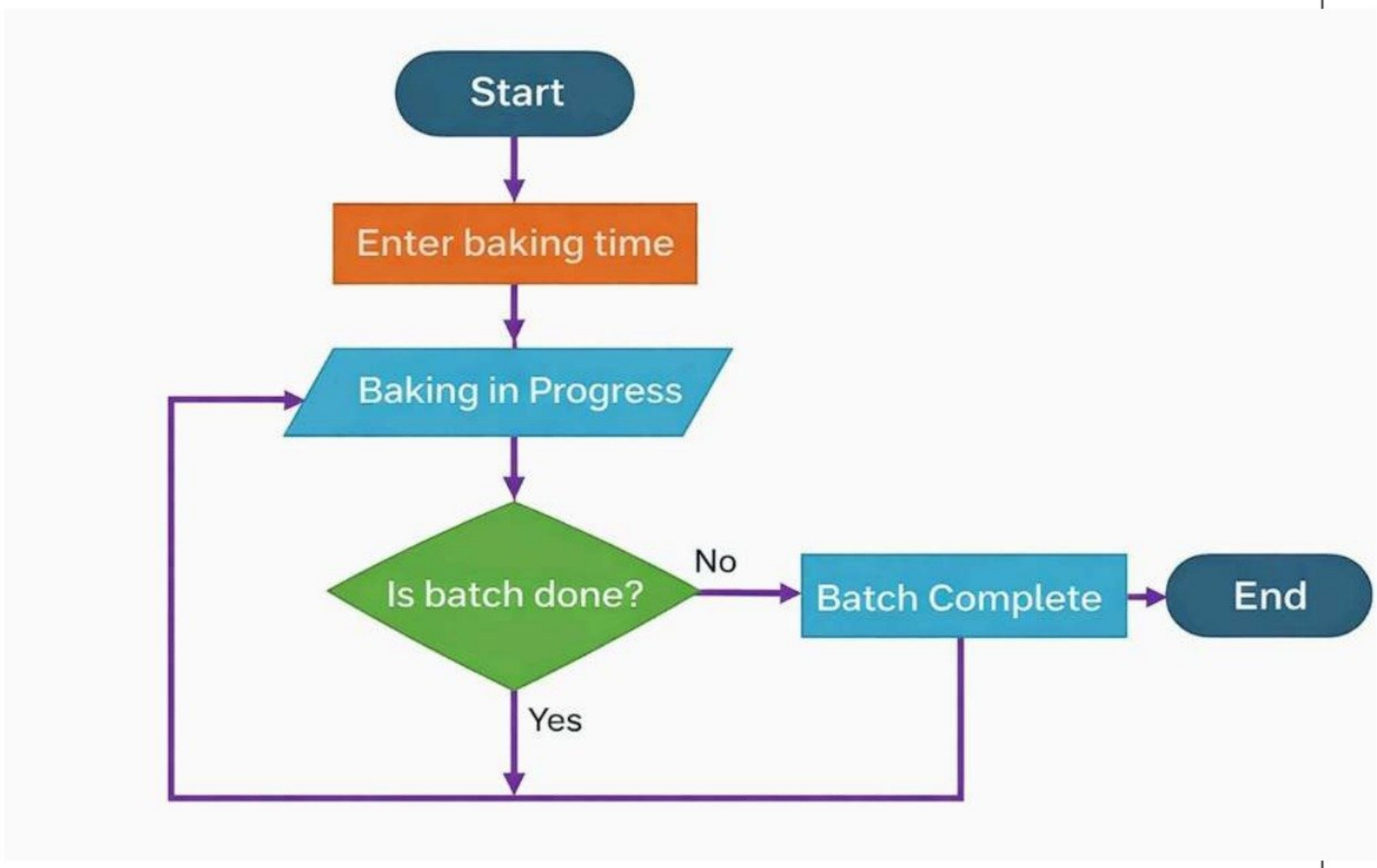
| 36. | A software developer is designing a simple interactive program that performs a countdown. The program works as follows:<br><br>• Ask the user if they want to launch a countdown.<br>• If the user replies **"yes"**, the program should perform a countdown from 100 to 1 (inclusive) by calling a **subprogram** named `countdown`.<br>• After completing the countdown, ask the user again whether they want to launch another countdown.<br>• Repeat this process until the user replies **"no"**.<br><br>Draw a **flowchart** representing the above scenario |
| 37. | A bakery uses an automated system to manage ovens. The system works as follows:<br><br>  <br><br>For a batch that requires 3 cycles to complete, indicate how many times the system checks the batch status and what is displayed in each check. |
| 38. | A printing system uses a flowchart to display numbers from 1 to 5 using a loop structure. The system repeats the printing process until the condition is satisfied.<br><br>Identify the **type of loop used** and the **number of iterations**. |
| 39. | A simple calculator system reads three numbers from the user and computes their total using a flowchart before displaying the result.<br><br>Identify the **input step** and the **processing step** used. |

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| 40. | A digital display system generates a special number sequence for a light pattern. The sequence is defined as follows:<br><br>• The first two numbers are **0 and 1**<br>• Each next number is the **sum of the previous two numbers**<br>• The system asks the user to enter the **number of terms (n)**<br>• If `n ≤ 0`, display **"Invalid Input"**<br>• Otherwise, generate and display the sequence up to `n` terms<br>• Also calculate and display the **sum of all generated numbers**<br><br>Draw a **flowchart** to represent the above system using a **loop**, clearly showing:<br><br>• Input and validation<br>• Initialization of first two terms<br>• Loop to generate the sequence<br>• Calculation of sum<br>• Output of sequence and total sum |
| 41. | An algorithm to find the factorial of a given number is given below<br><br>**Step 1:** Start the algorithm.<br>**Step 2:** Read the given number **n**.<br>**Step 3:** Set the value of **fact** to 1.<br>**Step 4:** Set the value of **i** to 1.<br>**Step 5:** Check whether **i** is less than or equal to **n**.<br>**Step 6:** Multiply **fact** by **i** and store the result in **fact**.<br>**Step 7:** Increase the value of **i** by 1.<br>**Step 8:** Repeat Steps 5 to 7 until **i** becomes greater than **n**.<br>**Step 9:** Display the value of **fact**.<br>**Step 10:** Stop the algorithm.<br><br>Draw an equivalent flowchart for the above algorithm. |

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| 42. | **The** Pseudocode to Check Whether a Number is Palindrome or Not is given below,<br>BEGIN<br>DECLARE n, temp, reverse, digit<br><br>INPUT n<br>temp ← n<br>reverse ← 0<br><br>WHILE temp > 0 DO<br>digit ← temp MOD 10<br>reverse ← (reverse × 10) + digit<br>temp ← temp DIV 10<br>END WHILE<br><br>IF reverse = n THEN<br>PRINT "Palindrome"<br>ELSE<br>PRINT "Not Palindrome"<br>END IF<br>END<br><br>Draw an equivalent flowchart for the above pseudocode. |
| | **Unit 2** |
| 1. | 1. A university system processes student records where the number of operations increases exactly in proportion to the number of students entered. For example, if 100 students are processed, about 100 operations are performed; if 1000 students are processed, about 1000 operations are performed.<br><br>Analyze the scenario and determine the time complexity of the algorithm. Justify your answer based on order of growth. |
| 2. | In a data processing application, Algorithm A has time complexity O(1), and Algorithm B has time complexity O(log n). However, Algorithm A involves a large constant number of operations, while Algorithm B has a very small constant factor.<br><br>Evaluate whether Algorithm A will always perform better than Algorithm B for all input sizes. Justify your answer using asymptotic analysis and practical performance considerations. |

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| 3. | 3.A hospital management system is designed to assign the nearest available ambulance to an emergency location. The system maintains a list of ambulances and checks their current positions to determine the closest one for faster response time.<br><br>Given the pseudocode to find the nearest ambulance, identify the key operation that influences the time complexity of the algorithm and justify your answer.<br><br>function find_nearest_ambulance(ambulances, emergency_location):<br><br>nearest_distance = INFINITY<br><br>nearest_ambulance = null<br><br>for each ambulance in ambulances:<br><br>distance = calculate_distance(ambulance.location, emergency_location)<br><br>if distance < nearest_distance:<br><br>nearest_distance = distance<br><br>nearest_ambulance = ambulance<br><br>return nearest_ambulance |
| 4. | A data processing system uses different algorithms to handle increasing input sizes. Some algorithms execute in constant time, while others show linear, quadratic, or exponential growth in the number of steps as the input size increases.<br><br>For a given input size of N = 100, select any three types of time complexities from O(1), O(log N), O(N), $O(N^2)$, and $O(2^n)$. Estimate the approximate number of steps involved for each and compare their performance. |

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| 5. | While developing a student management system for a university. The system maintains a sorted list of student records based on their student IDs. You need to quickly search for a student's details (e.g., name, department) using their unique student ID. Since the list is sorted, you can use binary search to efficiently find the student.<br><br>Python function to search for a student by their ID in a sorted list of student records using binary search is given below. Analyze the worst case, best case and average case time complexity for the given solution with justification.<br>def find_student_details(students, target_id):<br>left, right = 0, len(students) - 1<br>while left <= right:<br>mid = (left + right) // 2<br>mid_student = students[mid]<br>if mid_student['id'] == target_id:<br>return mid_student<br>elif mid_student['id'] < target_id:<br>left = mid + 1<br>else:<br>right = mid - 1<br>return "Student not found"<br><br>[Hint: Worst Case: The target is either not in the list or is found at the last step; Best Case: The target is found in the first comparison (middle element); Average Case: On average, the target is found after some 'n' steps] |
| 6. | A music streaming platform records the sequence of songs played by users during a session. The platform wants to identify the "most dominant song" in each session based on listening patterns.<br><br>Sequence 1: [rock, pop, rock, rock, jazz, rock]<br>Sequence 2: [pop, pop, rock, rock, rock, pop, pop]<br>Sequence 3: [jazz, classical, jazz, jazz, pop, jazz, jazz]<br><br>A song is considered "dominant" if it is played more than n/2 times in a sequence of length n. Decompose the problem and formulate an algorithm to determine the dominant song in each sequence by considering the following conditions:<br>a) If there is no dominant song, display the last played song as the answer<br>b) If there is a tie between two songs, the song that appears continuously (consecutively) should be the answer<br>c) If there is a tie and both songs appear continuously, then the answer should be "NIL" |
| 7. | A university operates a bus pass verification system to allow students to board campus buses. Each student has a unique ID, and all valid IDs are stored in an **unsorted list** containing nnn entries. Whenever a student attempts to board, the system scans the list **sequentially**, comparing the student's ID with each stored ID one by one until a match is found or the end of the list is reached. If the ID is found, the student is allowed to board; if it is not found, access is denied. Analyze the efficiency of this verification system and identify the **worst-case scenario** for checking a student ID, stating the exact number of comparisons made, and express the **worst-case time complexity** using **Big O notation**. |

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| 8. | A library membership system verifies a user ID by scanning a list of **n IDs sequentially**. If the user is the first in the list, the system finds it immediately. If the user is the last, it checks all n entries.<br><br>Describe the **best-case and worst-case scenario in terms of steps performed** without using Big O notation. |
| 9. | A smart parking system iterates through a list of **n available slots** to find the nearest slot to a driver's current location.<br><br>**Task:** Identify the **operation that dominates the processing time** and explain why it determines the system's efficiency. |
| 10. | An online shopping platform retrieves product details from a **sorted product list**. Method A scans products sequentially; Method B uses an index to jump directly to the product.<br><br>Compare the efficiency of Method A and Method B for large product lists and justify your reasoning. |
| 11. | A warehouse inventory system tracks **n items**. A robot picks items sequentially from a shelf to fulfill an order. If the required item is the first on the shelf, it's picked immediately; if it's last, the robot checks all n items.<br><br>Calculate the **average number of items the robot checks** to find a random item and explain your reasoning. |
| 12. | A treasure-hunting robot must find a hidden key inside a 5×5 grid of locked boxes. The robot opens boxes one by one in a sequential manner until it finds the key.<br><br>Analyze this search method by determining the maximum and average number of boxes the robot may need to open, explain why this approach is considered a brute-force method, suggest a strategy to reduce the number of checks, and discuss how increasing the grid size would affect the time required to find the key. |

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| 13. | A company is developing a program to place flags on a square grid for a display competition. The rules are:<br><br>1. No two flags can be in the same row.<br>2. No two flags can be in the same column.<br>3. No two flags can be on the same diagonal.<br><br>The program tries all possible arrangements one by one to find a valid placement. Currently, it works on an **8×8 grid**, but the company is considering larger grids, such as **10×10**.<br><br>Analyze the program in terms of:<br><br>• Maximum number of arrangements it may need to check.<br>• Average number of arrangements it will check before finding a solution.<br>• How increasing the grid size affects time and space required. |
| 14. | A text processing program searches for a pattern `p` of length `m` within a text `t` of length `n` using a nested loop approach. Analyze the algorithm to determine its **worst-case time complexity**, **space complexity**, and how changing the outer loop to `for (int i = 0; i < n; i++)` affects its performance.<br><br>for (int i = 0; i <= n - m; i++) {<br><br>int j = 0;<br><br>while (j < m && t[i+j] == p[j]) {<br><br>j++;<br><br>}<br><br>if (j == m) return i;<br><br>}<br><br>System.out.println("No match found");<br><br>return -1; |

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| 15. | A bookstore maintains records of its books. The records include **sorted book prices**. A program is used to search for a specific book price.<br><br>• **Book Prices (sorted):** `[200, 250, 300, 400, 500]`<br><br>Using **binary search** with an **iterative method**, find the book price `400`. What is the **space complexity** of this search and why? |
| 16. | A company maintains records of its employees. The records include **unsorted employee IDs** and **sorted employee salaries**. A program is used to search for specific employee details.<br><br>• **Employee IDs (unsorted):** `[110, 215, 320, 425, 530]`<br>• **Employee Salaries (sorted):** `[20000, 25000, 30000, 40000, 50000]`<br><br>Using **linear search**, find the employee ID `425`. What is the **time complexity** in the worst case and why? |
| 17. | A library has a **sorted list of book IDs**:<br><br>• **Book IDs:** `[101, 203, 305, 407, 509]`<br><br>If a librarian searches for the book ID `407` using **linear search**, it takes **4 steps**. Using **binary search** on the same sorted list, calculate how many steps it would take to find `407`. Show the steps of binary search and how the search space is reduced at each step. |
| 18. | A company has the following employee records:<br><br>• **Unsorted Employee IDs:** `[102, 215, 301, 410, 512]`<br><br>Use **linear search** to check whether employee ID `301` exists in the list. Write the steps taken and show how linear search checks each element one by one. |

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| 19. | Consider the below pseudocode<br><br>ALGORITHM PairSumBruteForce<br>SET numbers = [2, 7, 11, 15]<br>SET target_sum = 9<br>SET n = length(numbers)<br>FOR i FROM 0 TO n - 1 DO<br>FOR j FROM 0 TO n - 1 DO<br>IF i NOT EQUAL TO j THEN<br>SET current_sum = numbers[i] + numbers[j]<br>IF current_sum EQUALS target_sum THEN<br>PRINT "Pair found: " + numbers[i] + " and " + numbers[j]<br>RETURN<br>END IF<br>END IF<br>END FOR<br>END FOR<br>END ALGORITHM<br><br>Perform a complete technical evaluation of the PairSumBruteForce algorithm by quantifying its Time and Space Complexity in Big O notation based on the nested loop structure, calculating the exact factor by which the total operations increase if the input list size (n) is doubled from 500 to 1,000 elements, and identifying the specific efficiency classification within the hierarchy of computational growth. |
| 20. | A security firm is testing the strength of a high-security briefcase that uses a **3-digit combination lock**. Each digit can be any number from **0 to 5**. A "Brute Force Attack" involves a specialized machine trying every single possible combination until the lock opens.The machine takes **2 seconds** to attempt a single combination. |

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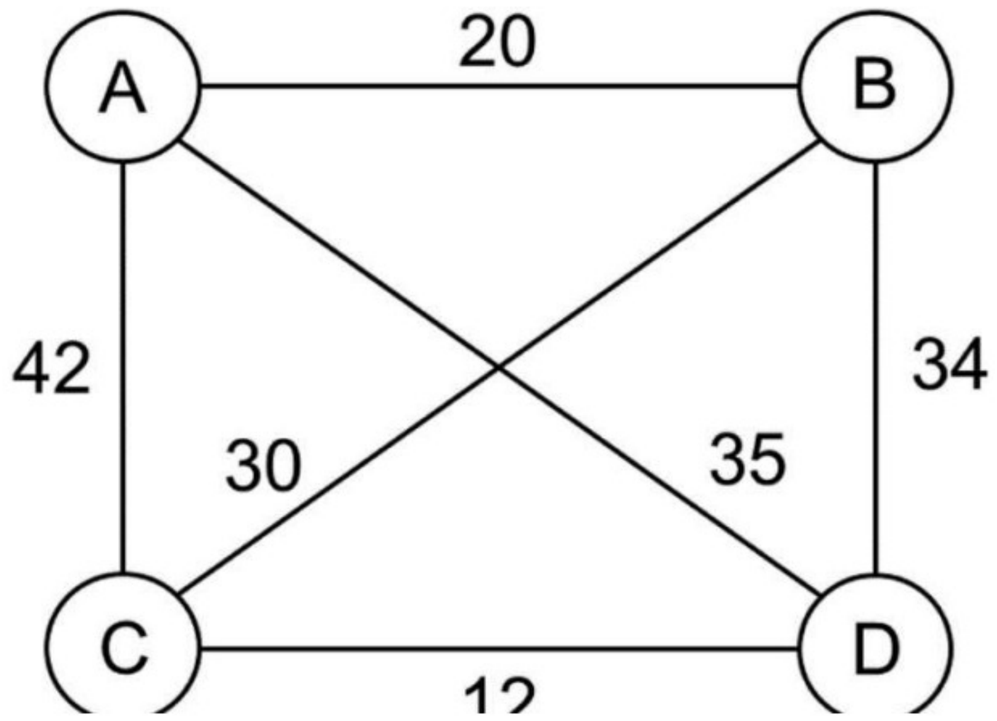
| 21. | A delivery service needs to deliver parcels to four locations: **A, B, C, and D**. The distances (in km) between the locations are shown in the graph:<br>  <br><br>Figure: Graph<br><br>The delivery person wants to start from location **A**, visit each location exactly once, and return to **A**. Using the **brute force approach**, determine all possible routes starting and ending at **A**, calculate the total distance for each route, identify the route(s) with the minimum total distance, and explain why this approach guarantees finding the shortest route. |
| 22. | A data processing system handles a large list of elements using a **divide and conquer strategy**. The system repeatedly splits the list into two equal parts at each step until smaller sublists are formed for processing.During execution, the system reaches a stage where there are **exactly 4 sublists**, each representing a portion of the original data.Indicate the **original number of elements in the list**. |
| 23. | A search engine needs to quickly locate a specific page number from a **sorted list of page IDs** containing millions of entries. To improve efficiency, the system repeatedly divides the list into halves and searches only the relevant part.<br><br>Identify the **algorithmic approach used** and indicate **why it is suitable for this application**. |
| 24. | A fitness tracking app records the **number of steps walked each day** over a period of time:<br>`[3000, 4500, 5200, 2800, 6100, 4000, 3500, 4800]`<br>To quickly calculate the **total number of steps**, the app uses a **divide and conquer approach** by splitting the list into smaller parts, computing partial sums, and combining them.<br><br>• Identify the **sum of elements** in the left and right sublists after the first division.<br>Identify the **final total number of steps**. |

25. A software company needs to find the **maximum sales value** from a large list of daily sales for the past week:

**Daily Sales (in $):**

```
[250, 420, 180, 560, 300, 410, 150]
```

The team decides to use a **divide and conquer approach** to find the maximum efficiently.

- Identify the **maximum sales value** using divide and conquer.
- Indicate how the original list can be **divided into sublists** for the first step.

26. A small e-commerce warehouse receives **unsorted product IDs** that need to be processed and shipped in ascending order. The warehouse staff uses **Quick Sort** to organize the IDs for efficient handling.

**Unsorted Product IDs:**

```
[50, 30, 70, 10, 60, 20, 40]
```

In an initial attempt, the staff chooses the **first element of the list as the pivot**. This choice results in **4 iterations (partitions)** to fully sort the list because the first pivot creates uneven partitions and requires multiple recursive steps to arrange the remaining sublists.

If the **middle element** of the list is chosen as pivot, identify the **number of iterations (partitions)** required to fully sort the list. Identify how the **pivot choice affects the number of iterations** in Quick Sort.

27. While developing an **inventory management system** for an electronics store, the system maintains a **sorted list of product records** based on their **product IDs**. You need to efficiently search for a product's details (e.g., name, category, price) using its unique product ID. Since the list is sorted, you can use **binary search** to find the product quickly.

**Python function to search for a product by its ID in a sorted list of product records using binary search:**

```
def find_product_details(products, target_id):
    left, right = 0, len(products) - 1
    while left <= right:
        mid = (left + right) // 2
        mid_product = products[mid]
        if mid_product['id'] == target_id:
            return mid_product
        elif mid_product['id'] < target_id:
            left = mid + 1
        else:
            right = mid - 1
    return "Product not found"
```

**Analyze the worst-case, best-case, and average-case time complexity** for the above function with justification.

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| 28. | A bookstore receives a shipment of books with different weights in grams. The staff needs to arrange the books on shelves so that lighter books are placed first and heavier books later for easy access. To do this efficiently, they decide to sort the book weights in **ascending order** using an efficient sorting algorithm.<br><br>**Book Weights Received Today:**<br><br>`[120, 250, 80, 300, 150, 200, 50]`<br><br>Sort the book weights using **Merge Sort** and indicate the **worst-case time complexity** of the algorithm. |
| 29. | A drone surveillance system is programmed to visit multiple checkpoints by always flying to the nearest unvisited checkpoint using a greedy strategy.<br><br>During execution, the drone reaches the final checkpoint, but the distance to return to the base station is unexpectedly high.<br><br>**Indicate the reason for the increase in total travel distance based on the greedy approach.** |
| 30. | A robot delivery system in a warehouse uses a greedy strategy to visit multiple storage points by always moving to the nearest unvisited location.<br><br>**Identify whether this strategy always guarantees the minimum total travel distance and indicate your answer with a reason.** |
| 31. | A tourist planning system helps a traveler visit multiple cities with minimum travel cost using a **greedy approach**. The traveler starts from City A and always chooses the nearest unvisited city at each step.<br><br>**Distance Matrix (in km):**<br><br>\| \| **A** \| **B** \| **C** \| **D** \|<br>\| A \| 0 \| 10 \| 15 \| 20 \|<br>\| B \| 10 \| 0 \| 35 \| 25 \|<br>\| C \| 15 \| 35 \| 0 \| 30 \|<br>\| D \| 20 \| 25 \| 30 \| 0 \|<br><br>Starting from City A, **identify the order of cities visited using the greedy method and indicate the total travel cost.** |

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| 32. | A delivery service is transporting valuable goods using a truck with a weight limit of 50 kg. There are multiple packages, each with a certain weight and value. The goal is to maximize the total value of the items loaded into the truck. The pseudocode for solving the above case is given below. With that, analyze the time complexity of the same.[Hint: Step 1: Let input size be 'n'; Step 2: Time complexity of sorting; Step 3: Time complexity for iterating through sorted list; Step 4: Overall time complexity]<br><br>BEGIN<br><br>INPUT: List of items with weight and value, Knapsack capacity W<br><br>SORT items by value-to-weight ratio in descending order<br><br>SET total_value = 0<br><br>SET current_weight = 0<br><br>FOR each item in sorted list:<br><br>IF current_weight + item.weight ≤ W THEN<br><br>ADD entire item to knapsack<br><br>total_value += item.value<br><br>current_weight += item.weight<br><br>ELSE<br><br>TAKE fraction of item to fill remaining capacity<br><br>total_value += (remaining_weight / item.weight) * item.value<br><br>BREAK // Knapsack is full<br><br>RETURN total_value<br><br>END |
| 33. | A fuel transportation company has a tanker with a limited capacity of 1000 liters. The company needs to transport different types of fuel, each with a specific volume and value per liter. The goal is to maximize the total value of fuel transported without exceeding the tanker's capacity. This is a classic example of the Fractional Knapsack Problem, where fractions of items can be included. Formulate an algorithm to fill the tanker with the maximum value of fuel without exceeding its capacity by considering the following conditions:<br>a)Fractions of fuel can be included (e.g., 50% of a fuel type).<br>b)The total volume of the selected fuels must not exceed the tanker's capacity.<br>c)The goal is to maximize the total value of the selected fuels. |

34. A cargo ship is loading high-priority containers with a maximum carrying capacity of **12 kg**. Each container has a specific weight and profit value. The goal is to maximize the total profit using the **Fractional Knapsack approach**.

| **Items** | 1 | 2 | 3 | 4 | 5 |
|---|---|---|---|---|---|
| **Weight (kg)** | 4 | 2 | 3 | 6 | 1 |
| **Profits** | 20 | 12 | 18 | 24 | 10 |

Step 1 is done for you. Complete:

- **Step 2:** Arrange items in descending order of Pi/Wi
- **Step 3:** Select items without exceeding capacity to maximize profit

## Step 1: Calculate Pi/Wi

| Items | 1 | 2 | 3 | 4 | 5 |
|---|---|---|---|---|---|
| Pi/Wi | 5 | 6 | 6 | 4 | 10 |

35. A space research team is loading scientific equipment into a shuttle with a maximum carrying capacity of **60 kg**. There are three types of equipment modules, each contributing different scientific value. To maximize the total research value, the team decides to use the **Fractional Knapsack approach**, since some equipment can be partially loaded.

| **Item (Equipment Type)** | **Weight (kg)** | **Value (Rs.)** | **Value/Weight (Rs./kg)** |
|---|---|---|---|
| Research Sensors | 15 | 90 | 6.0 |
| Data Modules | 25 | 125 | 5.0 |
| Power Units | 40 | 160 | 4.0 |

**Total Capacity of Shuttle: 60 kg**

Complete the steps 2, 3, and 4 given in the hint to find the amount of each equipment to be loaded to maximize value.

**[Hint: Step 1 already done]**
Research Sensors → 6.0
Data Modules → 5.0
Power Units → 4.0

36. A maze-solving system uses a backtracking approach to find a path from start to end. In one situation, the system explores all possible paths before finding the correct one.

**Identify whether this situation represents the best case, worst case, or average case.**

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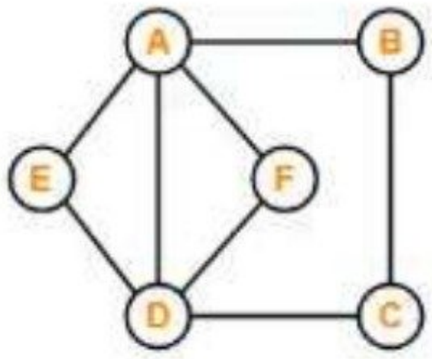
| 37. | A puzzle-solving application is designed to arrange pieces correctly. One method used by the system tries every possible arrangement of pieces, while another method uses a **backtracking approach** to avoid unnecessary attempts.<br><br>**Identify one key difference in how these two methods find solutions.** |
| 38. | A system uses a **backtracking approach** to generate all possible arrangements of the elements {A, B, C} without repetition.<br><br>During execution, the system forms the sequence:<br>A → AB → ABC<br><br>After generating this, the system removes C and continues the process.<br><br>**Identify why the system removes C after forming ABC and indicate the next arrangement generated.** |
| 39. | System generates all possible 3-digit codes using digits {1, 2, 3} without repeating any digit. It uses a **backtracking method** to build the code step by step.<br><br>While generating the codes, the system reaches the partial sequence **[1, 3]**.<br><br>**Identify why the system cannot directly complete the code from [1, 3] and indicate the next step performed using backtracking.** |

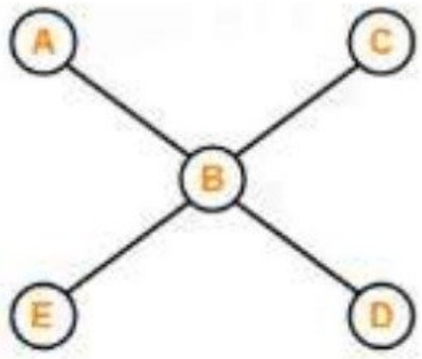
40. A network designer is analyzing different network topologies to determine whether a system can perform a **cyclic data transmission**, where each node is visited exactly once and the transmission returns to the starting node. This is modeled using the concept of a **Hamiltonian Cycle**.

Six different network graphs (A to F) are shown in the figure.

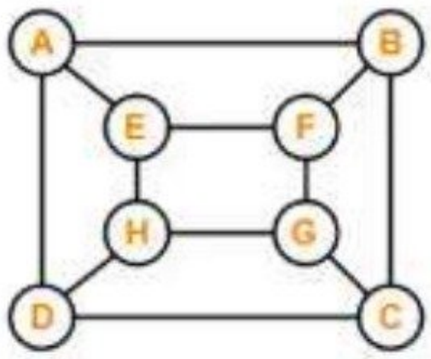
A)



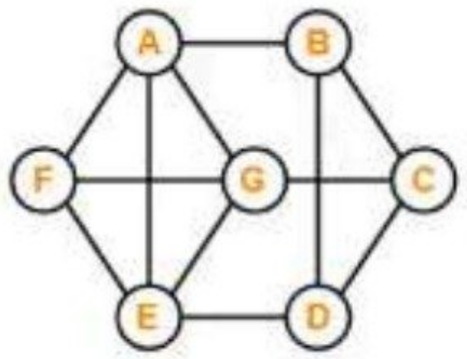

B)



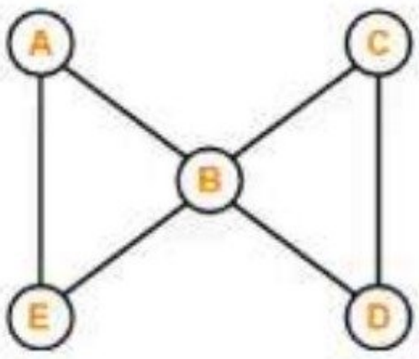

C)



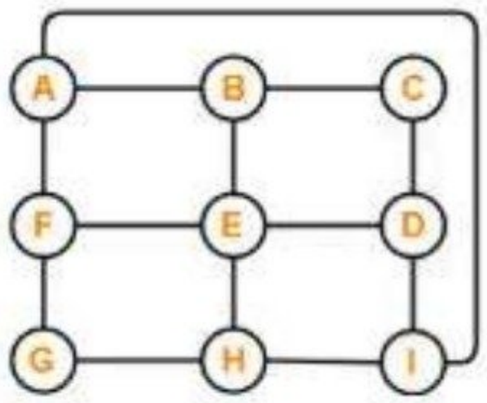

D)




E)



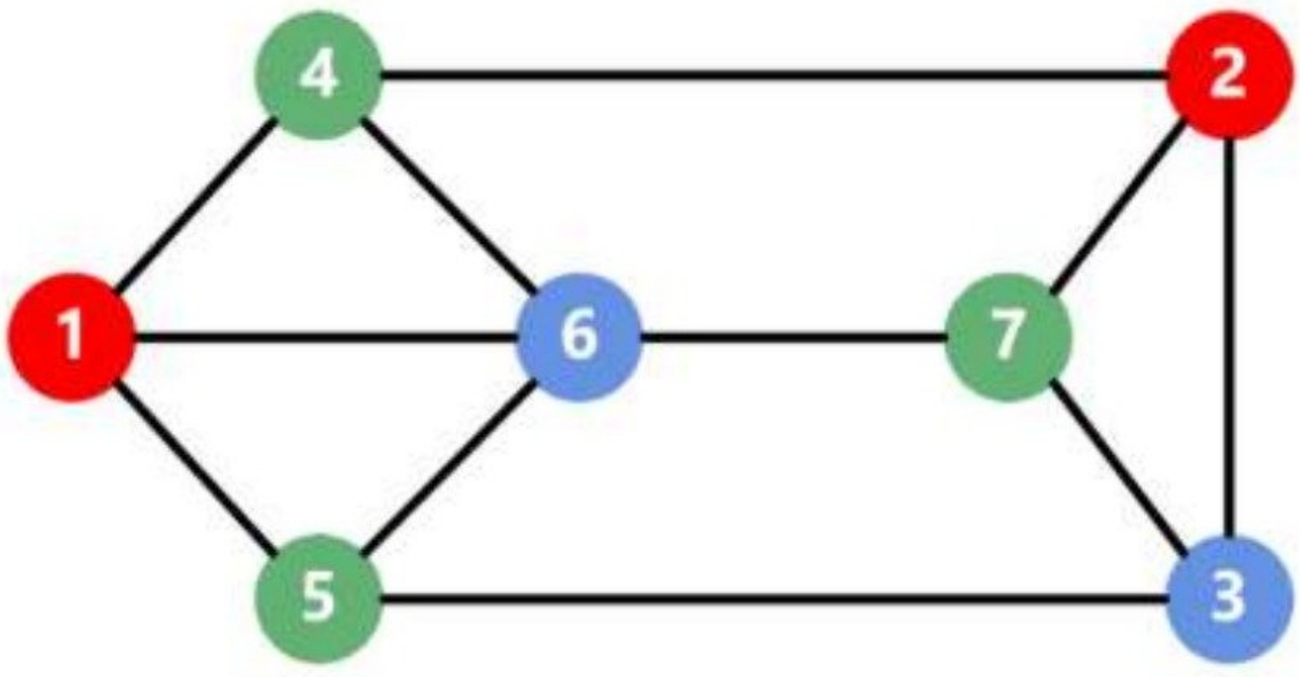

F)




**Identify which of the given graphs (A–F) contain a Hamiltonian Cycle and indicate one possible Hamiltonian cycle for any one valid graph.**

41. A network administrator is assigning different communication channels to a set of interconnected devices. Each device is represented as a vertex in the graph shown, and an edge between two vertices indicates that the devices are directly connected and cannot share the same channel due to interference.

The graph consists of **7 vertices (1 to 7)** connected as shown in the figure.




Using a **graph colouring technique**, assign colours (channels) to all the vertices such that no two adjacent vertices share the same colour. **Determine the minimum number of colours required and provide a valid colouring for the graph.**

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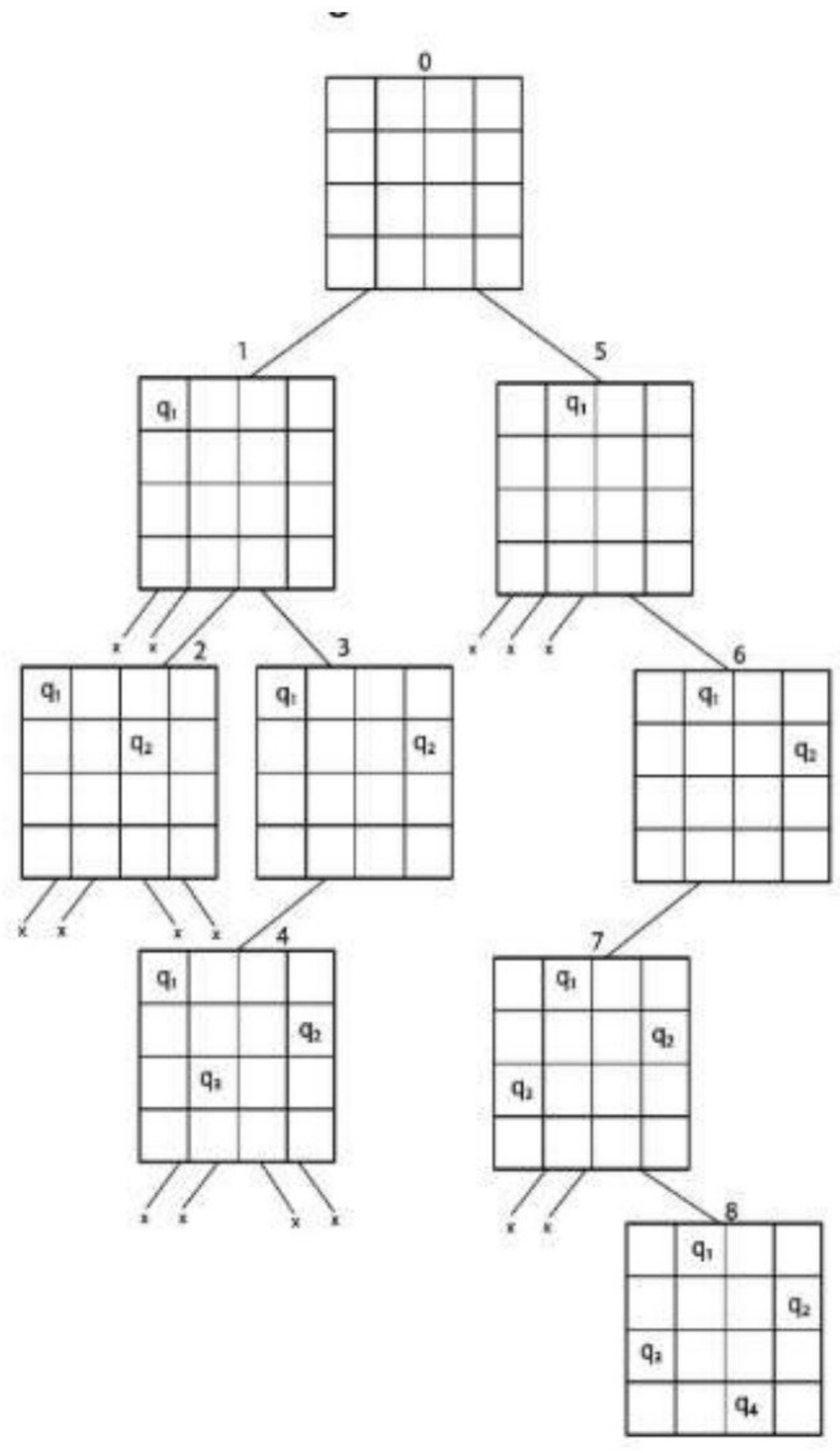
| 42. | Consider designing a wireless network system where multiple transmitters must be placed in a grid layout. Each transmitter is represented by a queen, and each grid position is represented by a cell on a chessboard. The goal is to place the transmitters such that no two interfere with each other (i.e., no two are in the same row, column, or diagonal).<br><br>Given the visual representation of a solution for a 4-transmitter setup in the figure titled *4-Queens*, provide any one valid arrangement for a **5-transmitter system** without interference.<br><br>  <br> |
| | **Unit 3** |

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| 1. | A social networking platform has millions of users. Each user can have multiple friends, and the platform needs to perform tasks like:<br><br>• Suggesting new friends based on mutual connections<br>• Quickly identifying friends of friends<br>• Displaying a network of connections for a user<br>• Finding communities or clusters of connected users<br><br>Each user is connected to other users in a complex network that doesn't follow a simple sequential order. The number of connections varies per user, and relationships are dynamic – new friendships are added and old ones removed regularly.<br><br>Identify the category of data structure suitable for representing these user relationships and justify. |
| 2. | A weather monitoring system collects daily readings for temperature, humidity, wind speed, and rainfall at multiple locations. Each reading is a single value recorded in the system's memory for further analysis and reporting.<br><br>Identify the most suitable type of data used to store these individual measurements. |
| 3. | A sales application stores customer information as comma-separated strings. Minor formatting changes cause runtime errors during string splitting and indexing. Developers propose reorganizing the data using structured records to eliminate dependency on string formats.<br><br>Assess how reorganizing program logic from delimiter-based strings to record-based data reduces errors and improves robustness. |
| 4. | A report-generation algorithm remains unchanged, yet its execution time drops significantly after reorganizing input data into structured records with logical grouping. This improvement surprises junior developers. Assess how data organization alone can enhance algorithm performance without modifying algorithmic steps. |

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| 5. | A university maintains a student management system where each student has a unique student ID, name, age, department, and GPA. The system stores all student records together to manage admissions, course enrollments, and exam results.<br><br>Identify the appropriate terms from elementary data organization that correspond to the following in this scenario:<br><br>• The single value representing a student's age<br>• The concept of a student in the system<br>• The characteristic "GPA" of a student<br>• The storage unit containing all details of one student<br>• The field that uniquely identifies each student record |
| 6. | An e-commerce website manages a large catalog of products. Each product has a **unique product ID**, **name**, **price**, **quantity in stock**, and **category**.<br><br>The website needs to:<br><br>• Store **individual values** such as price and quantity for performing calculations like total cost, discounts, and stock checks.<br>• Keep all product information **organized** so that:<br>  • Products can be **quickly searched** by ID or category<br>  • Products can be **sorted or filtered** by price or stock<br>  • New products can be **added**, and sold products **updated** dynamically<br>• Ensure the system can **scale efficiently** as the catalog grows to thousands of products.<br><br>Identify the appropriate **data types** and **data structures** for storing and managing the product information in this scenario and justify. |

| | |
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| 7. | When building any kind of project, it's essential to understand the clear distinction between data types and data structures. Data type refers to the kind or format of data that a variable can store. It defines the specific type of value that a variable can hold. Data structure, on the other hand, is a way of organizing and storing data in a program. It is a collection of different types of data, which can be grouped together and represented as a single unit or object.Justify with reasoning whether the following scenarios requires a data type or a data structure.<br><br>i) To store the price of an item in an e-commerce store, which can be a whole number like 100 or a decimal like 99.99.<br><br>ii) To store and retrieve pages in a web browser, where the last page visited is always accessed first.<br><br>iii) To store employee information in a tree, where each node contains data such as employee ID, name, and department, and has child nodes representing employees under the manager's supervision.<br><br>iv) To store a list of integers that represents the scores of 5 students in a class: [85, 90, 88, 92, 78]<br><br>Example:<br><br>i) To store a list of tasks in a to-do list app, where tasks can be added, removed, and marked as completed.<br><br>Data Structure – A to-do list app requires a way to organize and manage tasks, which is handled by a data structure (e.g., an array, linked list, or queue). |
| 8. | An online exam system ADT stores students' scores in a private array. It provides functions to add new scores and calculate averages. Users can retrieve the average but cannot see the raw scores.<br><br>Describe why users cannot access the raw scores directly and how they can still get results. |
| 9. | A warehouse management ADT keeps track of inventory counts. It provides functions to add items, remove items, and check stock. The way items are stored internally can be changed from arrays to linked lists without affecting the functions.<br><br>Identify what aspect of ADT allows the internal storage to change without affecting the functions. |

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| 10. | A restaurant ordering system is divided into separate modules: table management, order placement, and billing. Changing the billing logic does not affect table management or order placement.<br><br>State what happens to the other modules when the billing logic is updated and why the program remains manageable. |
| 11. | A **student record ADT** stores marks internally in a private array. Functions are provided to add, remove, and retrieve student marks. A student tries to directly access the array to modify marks but fails because the array is not visible outside the ADT.<br><br>Identify the ADT concept being applied here and justify your answer with reference to the scenario outcome. |
| 12. | Consider an Abstract Data Type (ADT) representing a **digital music playlist**, where each song has a title, artist, and duration. The ADT supports operations such as creating a playlist, adding new songs, deleting songs, updating song details, and checking information about the songs.<br><br>**Tasks:**<br>i) Identify and classify the operations described above into **create, copy, destroy, modification, and inquiry operations**. Explain the importance of **inquiry operations** in this ADT and why they do not change the playlist. |
| 13. | Consider an Abstract Data Type (ADT) representing a function to compute the **square of a non-negative integer**.<br><br>**Implementation in C:**<br><br>`#include <stdio.h>`<br><br>`int square(int n) {`<br>`  // Precondition: n >= 0`<br>`  return n * n;`<br>`  // Postcondition: Returns the square of n`<br>`}`<br><br>`int main() {`<br>`  printf("%d\n", square(-4));`<br>`  printf("%d\n", square(0));`<br>`  printf("%d\n", square(6));`<br>`  return 0;`<br>`}`<br>Identify which function calls in `main()` satisfy the precondition and which violate it.<br>Explain the postcondition for the function `square` and what it guarantees if the precondition is satisfied. |

14. Consider an Abstract Data Type (ADT) represented by an array of integers. The logical (abstract) view and implementation of the array are described as follows:

**Implementation view:**

- The array is of size 5, and its values are `{12, 25, 37, 48, 59}`.
- The index positions are 0, 1, 2, 3, 4, and each element is stored in contiguous memory locations.

**Logical (abstract) view:**

- The array stores a collection of integer elements.
- Allows access by index.
- Allows modification by index.
- Supports basic operations like summing, finding max/min, or sorting.

**Code Implementation:**

```
#include <stdio.h>

int main() {
    int arr[5] = {12, 25, 37, 48, 59};
    printf("%d\n", arr[1]);  // Output: 25
    arr[4] = 65;             // Modify the fifth element to 65
    return 0;
}
```

Predict the following:

i) Identify the type of operation performed when `arr[4] = 65` and explain how this relates to the concept of an Abstract Data Type.

ii) If the array is changed to a **dynamic array**, explain the impact on its abstract characteristics and whether it changes the logical view of the ADT.

15. A bookstore maintains a list of books in two different ways:

1. Using an array to store a fixed number of book IDs for a special collection of 50 books.
2. Using a linked list to store book IDs for general inventory, which keeps increasing as new books arrive.

The bookstore staff needs to:

- Quickly access the 20th book in the special collection.
- Frequently add new books to the general inventory.

For each storage method, identify whether it is a **static or dynamic data structure**

16. A hospital maintains a record of patient IDs in an array in the order of arrival:

```
patientIDs = [102, 115, 108, 120, 110];
```

The hospital staff performs the following actions:

1. They need to check all patient IDs to verify which patients have appointments today.
2. A new patient with ID `125` arrives and is added to the end of the array.

For each action, identify the type of array operation performed

17. A school keeps records of students and their marks using arrays in a C program. The data is organized as follows:

1. **Names of students in a single class** – e.g., `["Alice", "Bob", "Charlie", "David"]`
2. **Marks of students in 3 subjects** – e.g.,

```
[ [85, 90, 78],   // Alice
  [70, 88, 95],   // Bob
  [92, 80, 86],   // Charlie
  [60, 75, 82] ]  // David
```

1. **Monthly attendance of students across 4 classes and 30 days**

Decide which type of array (one-dimensional, two-dimensional, or multi-dimensional) should be used for each case and justify briefly.

18. Imagine you are managing a **to-do list application** where tasks are stored in an array in the order they were added. You can add, remove, access tasks, and search for a specific task.

**Current To-Do List:**

```
EMAIL CLIENT  COMPLETE REPORT  MEETING PREP  CALL SUPPLIER
```

Assess which of the following operations would take the longest time with reasoning:

1. Accessing the task "`MEETING PREP`" by its index.
2. Inserting a new task at the beginning of the list (e.g., "`SUBMIT INVOICE`").
3. Deleting the task at the beginning of the list (e.g., "`EMAIL CLIENT`").
4. Searching for the task "`COMPLETE REPORT`" using linear search.

| | |
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| 19. | A hospital maintains a **singly linked list** to keep track of patient records. Each node stores the **Patient ID** and **Admission Date**. The operations on the list are described below:<br><br>• Accessing the 5th patient record requires traversing the list.<br>• Searching for a patient by ID involves checking each node sequentially.<br>• Adding a new patient at the beginning of the list inserts the node immediately.<br>• Removing the first patient from the list deletes the node directly.<br><br>**Tasks:**<br>i) Determine the **time complexity (Big O notation)** for each of the following operations:<br><br>• Accessing a patient record at a specific position<br>• Searching for a patient by ID<br>• Adding a patient at the beginning<br>• Deleting the first patient<br><br>ii) Justify why these operations have the given complexities based on the linked list structure. |
| 20. | A library system keeps track of borrowed books using a linked list. Each node contains the book ID and student ID. The current list of borrowed books is:<br><br>`[BookID: 201, StudentID: 11] → [BookID: 202, StudentID: 12] → [BookID: 203, StudentID: 13]`<br><br>Operations performed:<br><br>1. A new book **BookID: 204** borrowed by **StudentID: 14** is added at the end.<br>2. The book with **BookID: 202** is returned and removed from the list.<br>3. Display the list of currently borrowed books.<br><br>**Tasks:**<br>i) Draw or write the linked list after performing the operations. (3 marks)<br>ii) State the final sequence of Book IDs in order. (2 marks) |
| 21. | Consider an application: Music Playlist – Storing the names of songs in a playlist. The choice of array for the above given application can be One-Dimensional array as it stores a sequential list of song names.Representation of One-Dimensional array: songs = ["Song A", "Song B", "Song C", "Song D"] Justify with reasoning the choice of array type and its representation that is required for the following application scenarios:<br><br>Application 1: Cinema Seat Booking System – Storing seat availability in a cinema hall.<br><br>Application 2: Weather Simulation Model – Storing temperature, humidity, and wind speed at different altitudes, locations, and times. |

| | |
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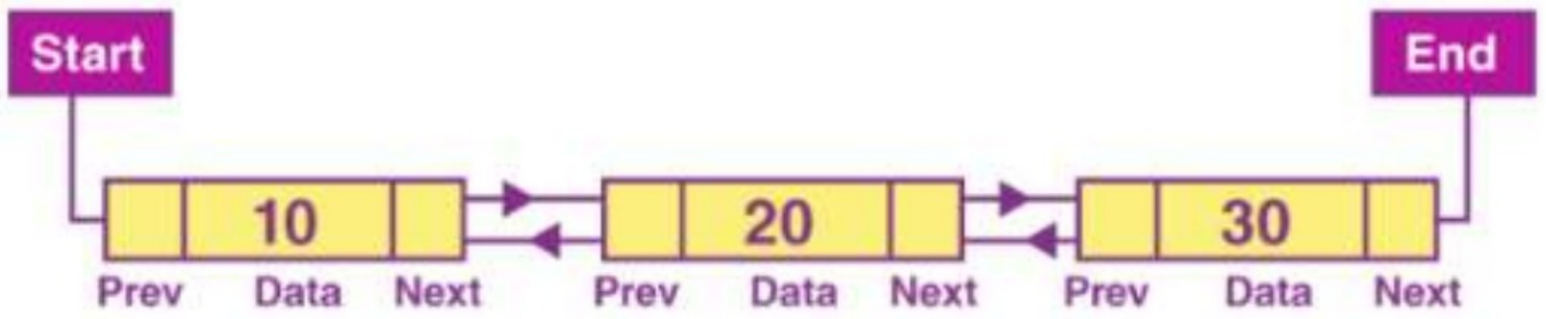
| 22. | A library uses a **stack** to manage returned books. The most recently returned book is placed on top of the stack.<br><br>State which **stack operation** is performed when a librarian adds a newly returned book and briefly describe its effect. |
| 23. | A hostel maintains a **singly linked list** of room numbers to track occupied rooms. When a new student checks in, their room number is added to the **end of the list**.<br><br>State which **linked list operation** is performed when a new room number is added at the end and briefly describe what it does. |
| 24. | A bus stop maintains a queue for passengers waiting to board. Initially, the queue contains the following passenger IDs in order of arrival (front → rear):<br>`[201, 202, 203, 204]`<br>The following actions occur:<br>1. A new passenger with ID `205` arrives and joins the queue.<br>2. The first passenger in the queue boards the bus and leaves.<br><br>Determine the **passenger at the front of the queue** and **list all passenger IDs in the queue** after performing the above operations. |
| 25. | Consider four different real-world situations where tasks or items need to be processed in a sequence. Predict which type of queue would be most suitable and justify briefly.<br>1. In a **fast-food restaurant drive-thru**, cars line up to place their orders. The first car to arrive is the first car to be served, and cars move forward as orders are completed.<br>2. In a **library book drop-off system**, books can be returned either at the main entrance or at the back entrance, depending on where the librarian is stationed.<br>3. In a **video streaming service**, multiple videos are queued to pre-load. When the last video finishes pre-loading, the system cycles back to the first video to continue buffering, ensuring smooth playback.<br>4. In a **fire station emergency call system**, calls are attended based on the urgency of the situation rather than the time the call was received. The most critical emergencies are handled first.<br><br>1. Example : **Simple Queue:** Cars are served in the order they arrive, following the FIFO principle. (1 mark) |

26.

1. You are designing the back-end logic for a simple text editor's 'Undo' buffer. Each time a user performs a major action, like typing a sentence, the state of the document is stored. Because users need to navigate backward and forward through their edit history, you have implemented this as a **doubly linked list** called `EditHistory`.

The current state of `EditHistory` is depicted in **Figure 1**:

- **Start (Head):** Points to the very first saved state.
- **End :** Points to the most recent saved state.
- **Nodes [10], [20], [30]:** Represent three progressive versions of the text document.




You are tasked with resolving the final visual structure of the `EditHistory` linked list after completing three sequential operations."

Based on Doubly linked list , execute the following three operations in exact sequence. **Depict the final structure of the `EditHistory` doubly linked list *after* all three operations are complete.** You must show the correct placement of all remaining data nodes and the proper pointers for `Start`, `End`, `Prev`, and `Next`.

1. **Delete from Beginning:** The oldest historical record is determined to be irrelevant and is removed from the buffer.
2. **Delete from End:** The most recent edit is corrupted and must be rolled back (removed).

**Insert at End:** A new corrected document state, **Data Node [40]**, is generated and added as the new most recent state.

27.

A web browser uses stacks to manage the Back and Forward navigation system.

- **Back Stack (S1):** Stores previously visited webpages.
- **Forward Stack (S2):** Stores pages that can be revisited after going back.

Initially, the user has visited the following pages in order: **Home**, **News**, **Blog**.

The user now visits a new website: **StackOverflow**. After that, the user presses **Back twice**.

Update the Back Stack (S1) and Forward Stack (S2) after each operation. Conclude the **current page**, **top of the Back Stack**, and **contents of the Forward Stack**.

28. You are managing a library system that tracks books being borrowed (using a stack) and books being returned (using a queue). Each book is referenced by its book ID.

**Borrowing Process (Stack) Operations:**

- `push(101)`
- `push(103)`
- `pop()`
- `push(107)`
- `push(110)`
- `s = pop()`

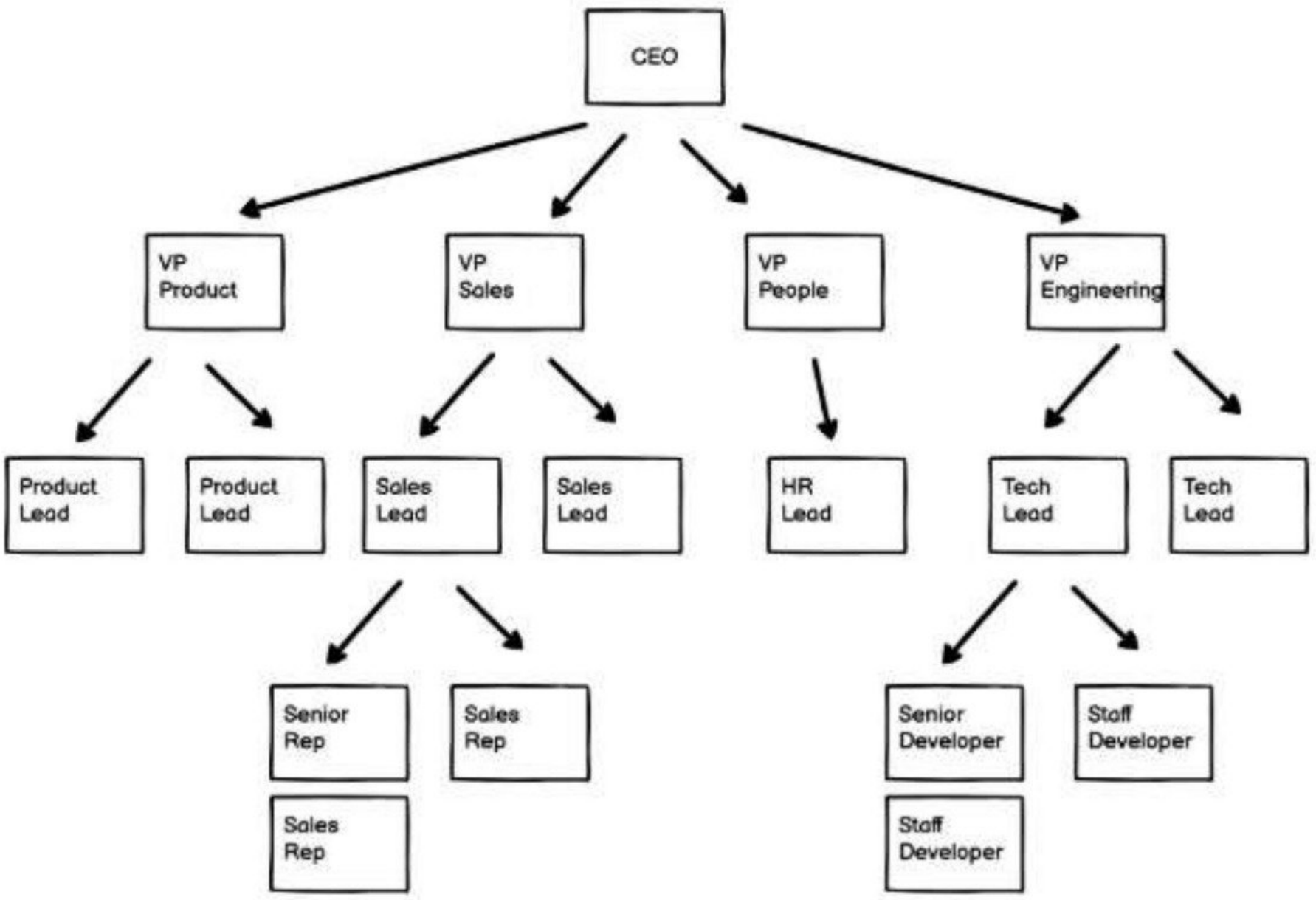
**Returning Process (Queue) Operations:**

- `enqueue(201)`
- `enqueue(205)`
- `dequeue()`
- `enqueue(208)`
- `enqueue(212)`
- `q = dequeue()`

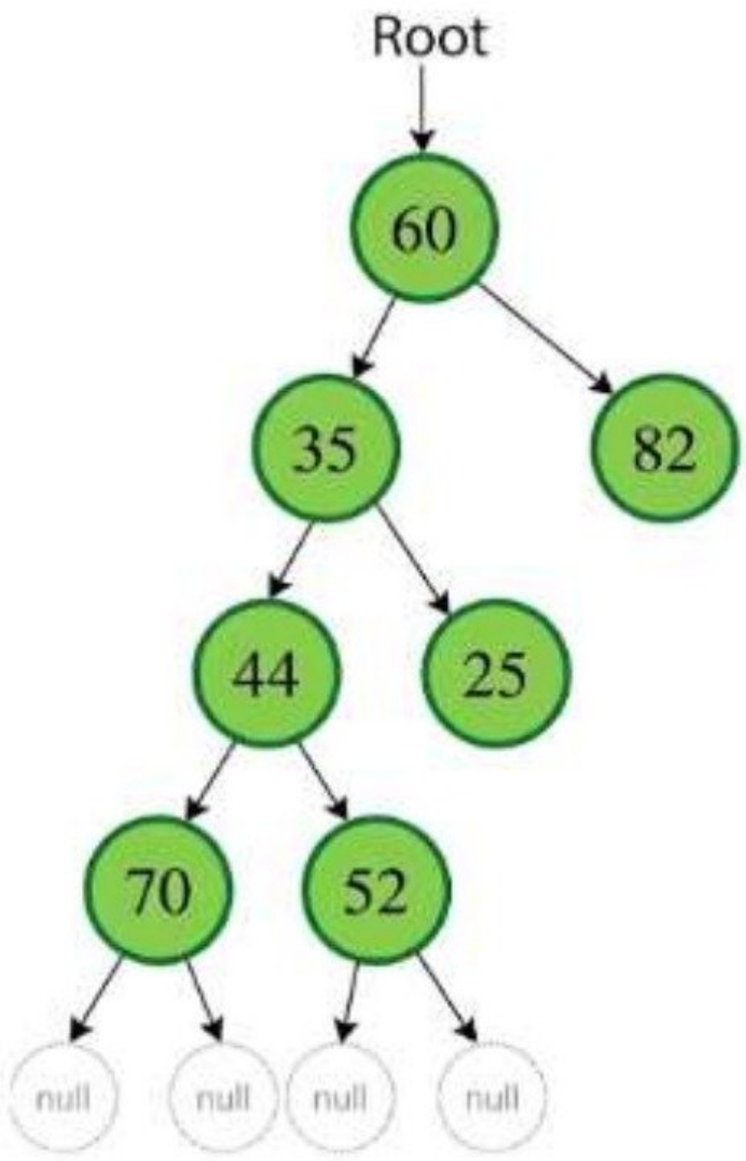
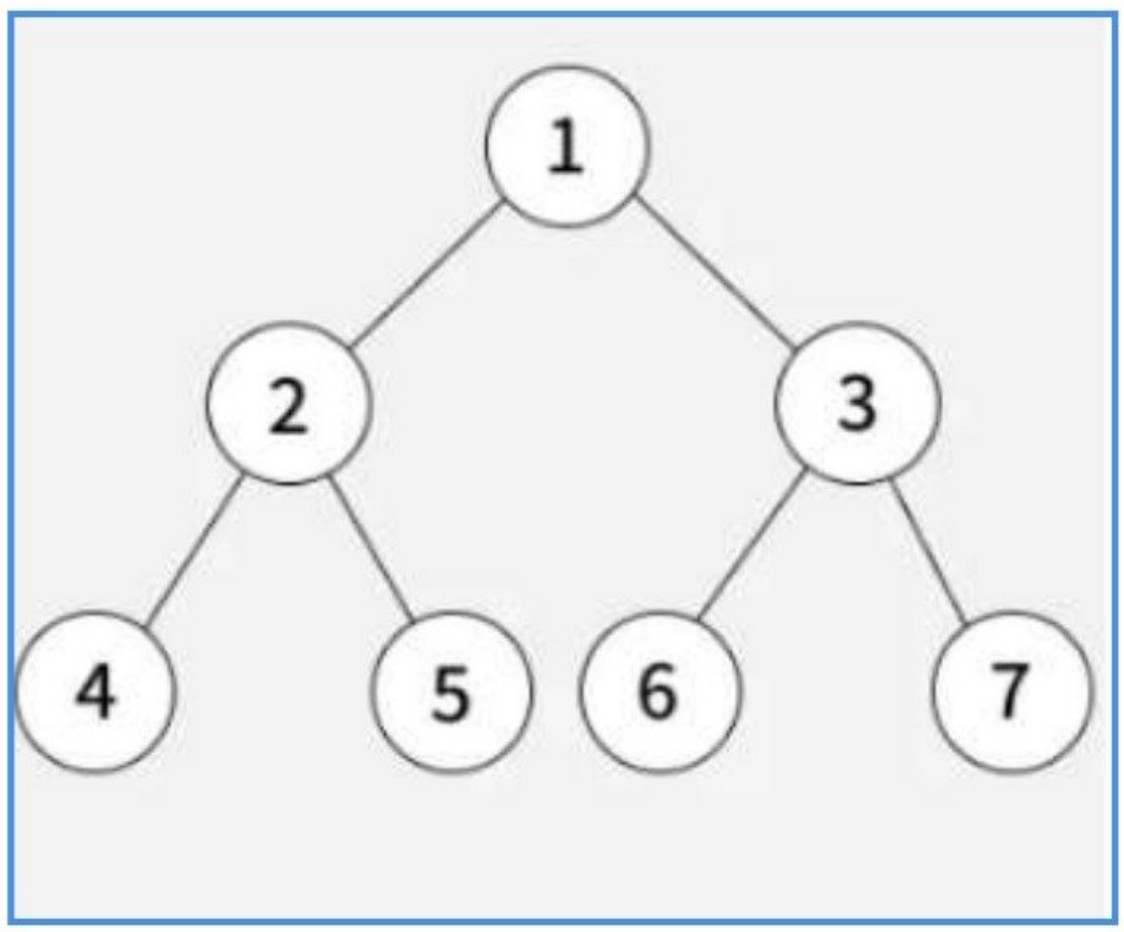
Compute the value of `stack + queue` step by step, which represents the total of the IDs of the books that were last borrowed and last processed for return.

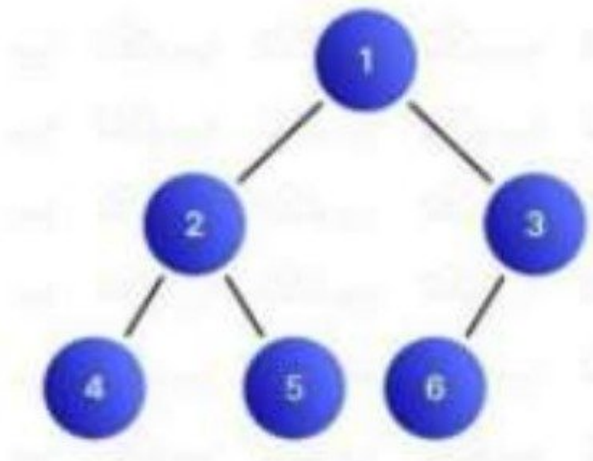
29.

Consider a company's organizational hierarchy represented as a tree, where the CEO is at the root, and each subsequent level of employees is arranged as branches. The root (CEO) has direct reports (like department heads), and each department head manages a set of employees.

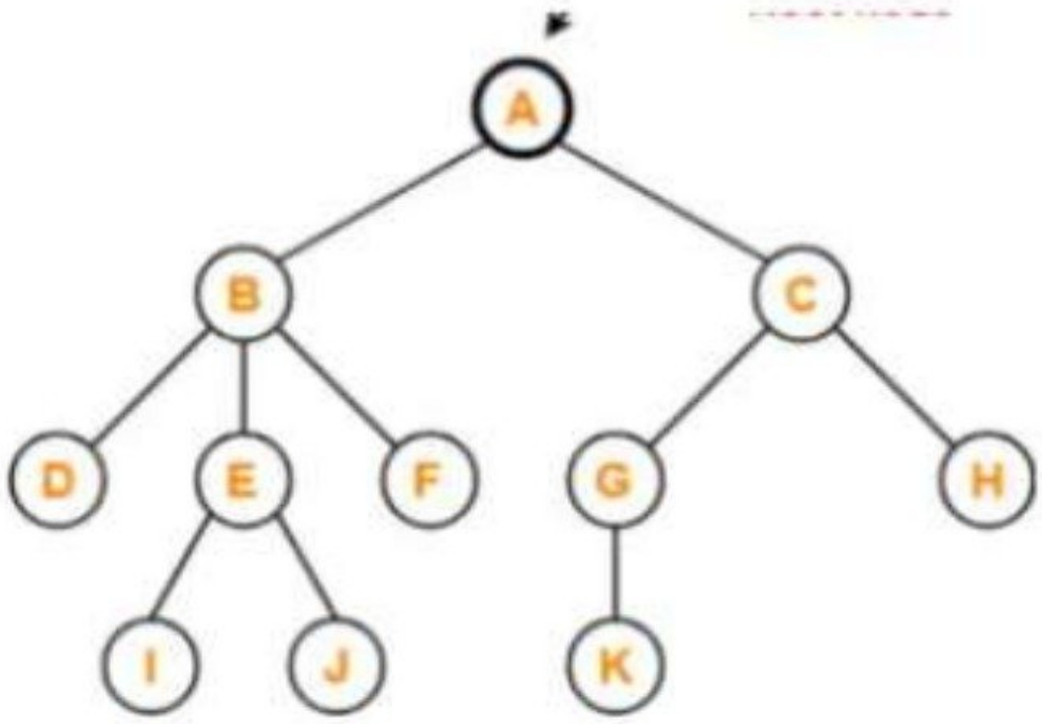



Construct a similar tree to represent a university's academic structure in a proper hierarchy.

| | |
|---|---|
| 30. | 1. A **Smart City management system** uses a network of acoustic sensors to monitor traffic density (measured in decibels/vehicle count). To ensure the central database remains balanced for quick searching, new sensor data is inserted into a **Binary Tree** using **Level-Order Insertion** (filling each level from left to right).**Initial State:** The first sensor reading, **60**, is recorded and established as the **Root** of the binary tree.**New Sensor Readings (in order of arrival):** `35, 82, 44, 25, 95, 18, 70, 52`<br>  <br>Figure titled final binary tree depicts the binary tree after inserting all the sensor readings, however there is an error in this tree. |
| 31. | CONSIDER the below tree<br>  <br>And identify the inorder ,pre-order and post-order |

| | |
|---|---|
| 32. | Convert the above complete binary tree to Balanced Binary Tree by adding or deleting a node   |
| 33. | A vending machine works as follows:<br>• Root: Insert Coin<br>• Branch 1: Choose Drink → Soda, Juice<br>• Branch 2: Choose Snack → Chips, Candy<br><br>Identify the **possible outcomes** after inserting a coin. |
| 34. | A project has tasks with dependencies:<br>• Task A must be completed before Task B and Task C.<br>• Task B must be completed before Task D.<br>• Task C must be completed before Task E.<br><br>If you represent this as a dependency tree, identify the **tasks that can be started immediately** (i.e., have no dependencies). |

35. Imagine a company's organizational chart, where the CEO is at the root of the hierarchy. The CEO oversees multiple Department Heads (Level 1). Each Department Head manages several Managers (Level 2), and each Manager supervises a team of Employees (Level 3). The structure helps visualize the reporting relationships within the company.Consider the following tree representing an organizational chart with dummy alphabet representation:




Infer the following from the given Tree:

1. Child Node of "D"

2. Sibling of node "E" & "F"

3. Leaf nodes

4. Nodes at Level 2

5.Number of Edges in the tree

36.

Consider a delivery network where different cities are connected by various transportation routes, such as highways, railways, or flights. Each city is represented as a node, and each direct route between two cities is represented as an edge. The weight of each edge represents the time (in hours) required to travel between two cities.

Cities:

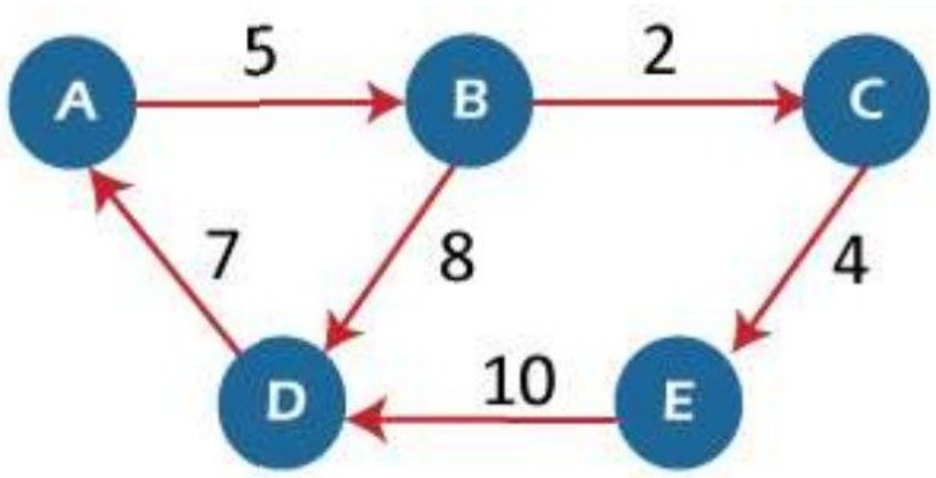
A - Delhi

B - Mumbai

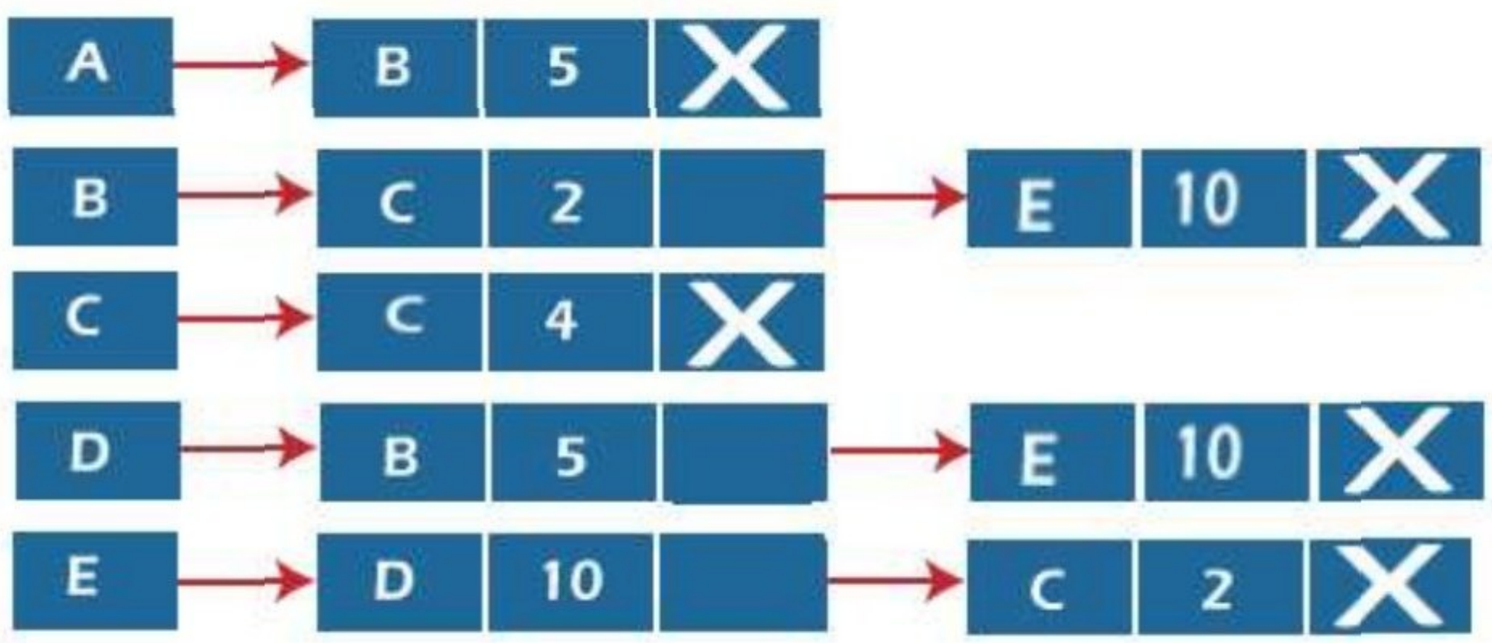
C - Bangalore

D - Chennai

E – Kolkata




Spot the error and correct it in the below given figure titled Adjacency List.



37. Consider a flight network where airports are connected by direct flights. Each airport is represented as a node, and each direct flight between two airports is represented as an edge. The network can be represented as a graph where the adjacency matrix provides a convenient way to store and analyze the connections between airports.

A- Delhi Indira Gandhi Airport

B- Mumbai Chhatrapati Shivaji Airport

C- Bangalore Kempegowda Airport

D- Chennai International Airport

E- Kolkata Subhas Chandra Bose International Airport

Figure titled "Graph Representation of Flight Network" represents the Graph of Flight Network of the airports are connected by direct flights.

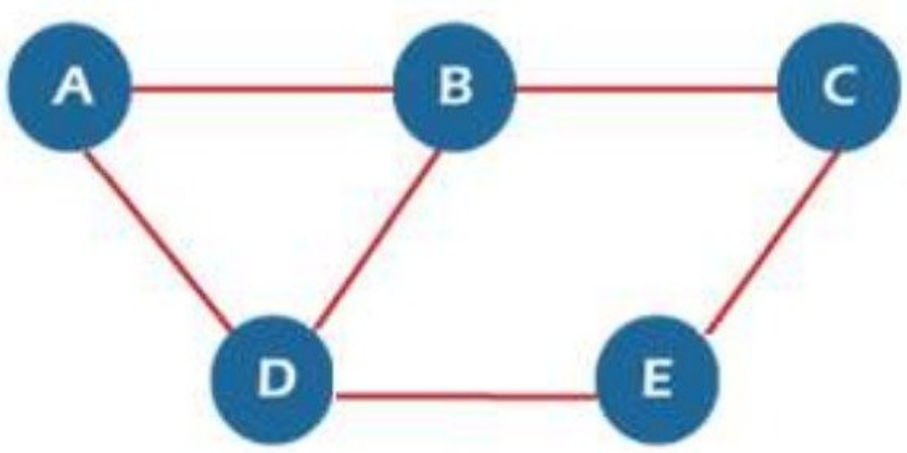



Figure: graph representation

i) Complete the respective adjacency matrix of the flight network given in the below figure titled "Adjacency Matrix of Flight Network".

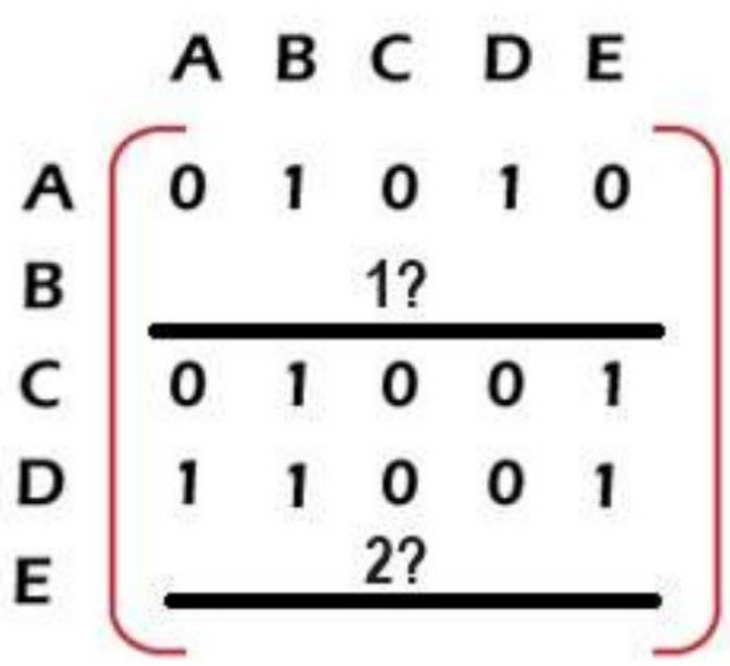
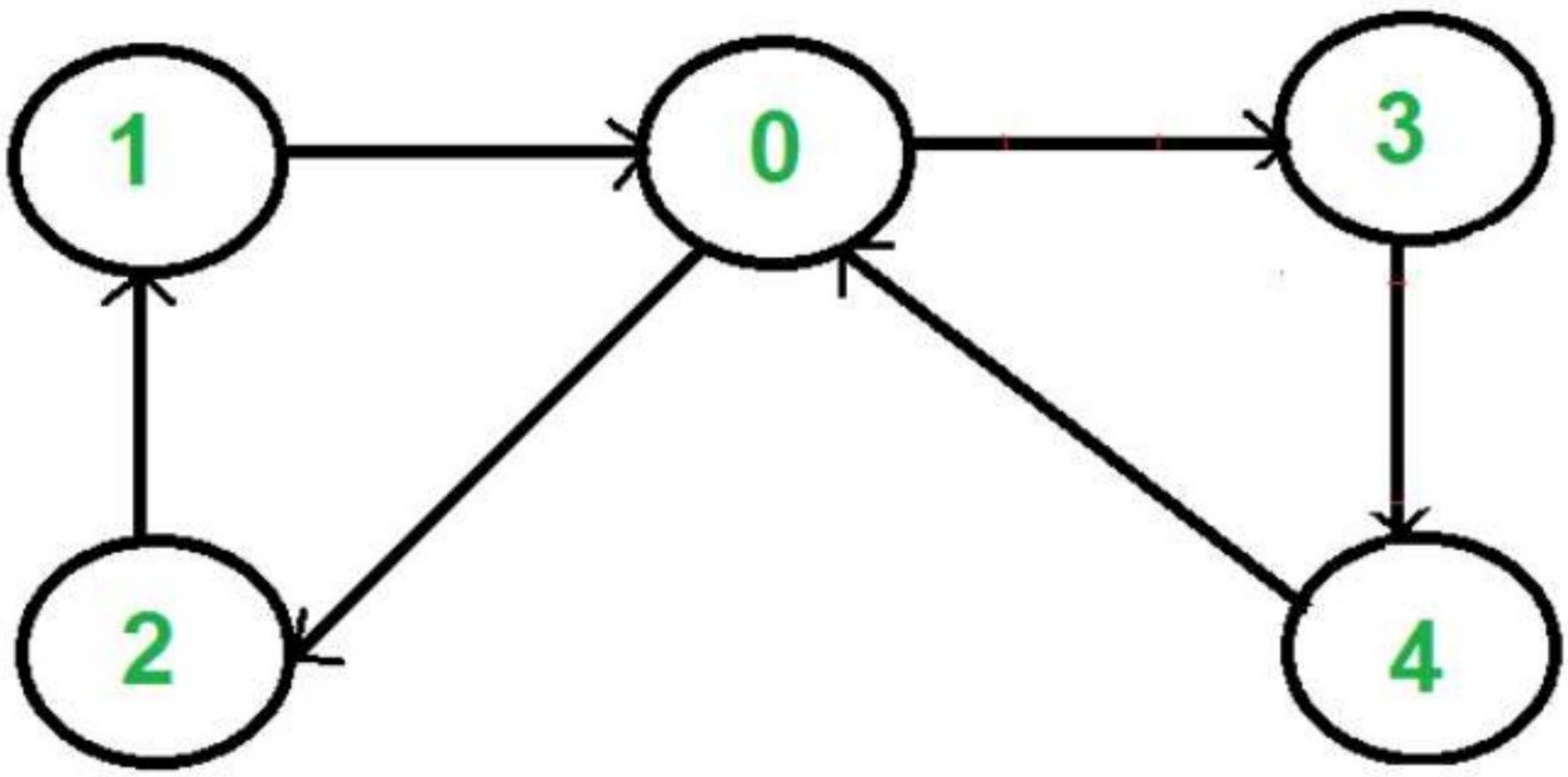



Figure: Adjacency Matrix of Flight Network

| | |
|---|---|
| | ii) If suppose the adjacency matrix value for C turns as C: 0 1 0 1 0, redraw the corresponding graph representation. |
| 38. | A **metro rail system** maps stations as nodes and tracks between stations as edges. Some stations allow trains to travel in both directions, while others are one-way.<br><br>Identify the type of graph used for:<br>a) Two-way travel only<br>b) One-way travel |
| 39. | Identify whether the given graph forms a cycle or not justify your answer<br><br>  |
| 40. | A **veterinary clinic** maintains a doubly linked list to manage the queue of pets for check-ups. Initially, the queue contains:<br><br>**Pet X (09:00 AM) -- Pet Y (09:30 AM) -- Pet Z (10:00 AM) -- Pet W (10:30 AM)**<br><br>**Operations performed:**<br><br>1. Insert Pet V (09:15 AM) before Pet Y.<br>2. Remove the first pet (Pet X).<br>3. Update Pet Z's appointment to 10:15 AM.<br><br>Predict the **third position** in the queue by listing the queue contents step by step after performing these operations. |

| | |
|---|---|
| 41. | Imagine a small town park with different zones, and walking paths connect these zones. The zones are represented as nodes, and the paths between them are represented as edges.<br><br>• Z1 is the central pavilion.<br>• Z2, Z3, Z4, and Z5 are surrounding zones.<br>• Connections:<br>  • Z1 is connected to Z2 and Z4<br>  • Z2 is connected to Z1 and Z3<br>  • Z3 is connected to Z2 and Z5<br>  • Z4 is connected to Z1 and Z5<br>  • Z5 is connected to Z3 and Z4<br><br>Assess which of the following properties of the park path network are violated and give the reason:<br><br>1. Self-loops: A path that connects a zone to itself.<br>2. Multiple paths: More than one path connecting the same pair of zones.<br>3. Connectedness: All zones are reachable from any other zone.<br>4. Planarity: The path map can be drawn on a piece of paper without any paths crossing each other. |
| 42. | A **cargo loading facility** monitors vehicles entering the warehouse. Based on the number of wheels, the system assigns a **category code**:<br><br>**Rules:**<br><br>• 2 wheels → 1 (e.g., bicycles, motorcycles)<br>• 4 wheels → 2 (e.g., small delivery vans)<br>• More than 4 wheels → 3 (e.g., trucks, forklifts)<br>• Otherwise → 0 (invalid inputs)<br><br>**Input:**<br><br>`vehicles = [2, 4, 8, 3, 12, 1]`<br><br>**Output:**<br><br>`newArray = [1, 2, 3, 0, 3, 0]`<br><br>Construct an algorithm to generate a **new array** where each vehicle's wheel count is replaced by the corresponding category code. |
| | **Unit 4** |
| 1. | A small library maintains a list of books with details like title, author, and price in a single file. The system handles only a small amount of data and does not require linking between different records.<br><br>Explain why a flat-file database is an appropriate choice for this scenario. |

| | |
|---|---|
| 2. | A small tuition center maintains student details like name, phone number, and marks in a single Excel sheet. There are no separate tables or relationships.<br><br>Identify the type of database used and mention one key characteristic. |
| 3. | A school stores student data in a flat-file database. The system allows only basic search operations and does not support advanced queries. Duplicate records are often entered without any validation.<br><br>Examine the characteristics of the system and explain how they affect data management. |
| 4. | An organization manages different types of data for its operations. Sales records are stored in tables, web data is collected in XML format, and emails and images are stored separately. These data types are used for reporting, communication, and content management.<br><br>Classify the data given in the scenario into appropriate type with justification |
| 5. | A small business stores its sales data in CSV files for easy sharing across systems. The system works well initially without requiring any special software. However, as the volume of data increases, managing and processing the data becomes difficult.<br><br>a) Identify any two characteristics of the flat-file database evident from the scenario.<br><br>b) Explain the effects of these characteristics on data handling.<br><br>c) State whether this system is suitable for large-scale use and justify your answer. |
| 6. | A small e-commerce business stores product and order details in a single flat-file database. As the number of customers and orders increases, the system becomes difficult to manage. They are unable to efficiently link customer orders with corresponding product information.<br><br>a) Identify the issue related to the database structure.<br>b) Explain the reason for this issue in a flat-file system.<br>c) Suggest a suitable solution with justification. |
| 7. | A mid-sized organization maintains employee and payroll data using a traditional file processing system. Employee details such as name, department, and salary are stored in multiple files maintained by different departments. When updates are made, inconsistencies often occur between files. Users frequently find it difficult to generate reports, as separate programs must be written for each query. Multiple employees sometimes update the same records simultaneously, leading to incorrect data entries. In addition, the system does not have proper access control, allowing unauthorized users to view sensitive salary information. During system crashes, incomplete updates leave the data in an inconsistent state.<br><br>Identify any **five disadvantages** of the file processing system and **justify each with reference to the scenario**. |

8. A database system stores employee details where age must be an integer, name must be a string, and joining date must follow a valid date format. The system rejects any values that do not match the expected data type.

State the type of constraint applied in this scenario and explain its purpose.

9. A student database stores information such as ID, Name, Age, and Course in a table as shown below:

| ID | Name | Age | Course |
|---|---|---|---|
| 1 | Ajeet | 24 | B.Tech |
| 2 | Aryan | 20 | C.A |
| 3 | Mahesh | 21 | BCA |
| 4 | Ratan | 22 | MCA |
| 5 | Vimal | 26 | BSC |

Examine the given table and determine:
a) The **degree** of the relation
b) The **cardinality** of the relation

10. Consider the two relations given below:

**R(A, B)**
(1, X), (2, Y), (3, Z)

**S(A, B)**
(2, Y), (3, Z), (4, W)

Write the appropriate **relational algebra expressions** to perform the following operations and determine their results:a) Union of R and S

b) Intersection of R and S
c) Difference of R and S

11. A banking system processes money transfers between accounts. During a transaction, the system ensures that either the entire amount is transferred successfully or no change occurs. It also maintains correct account balances, prevents interference from other transactions, and ensures that completed transactions are saved permanently even after a system failure.

Demonstrate the role of any 3 ACID properties in maintaining transaction reliability and data consistency.

12. Consider two tables, STUDENT and BRANCH.

**STUDENT Table**

| ROLL_NO | NAME | BRANCH_CODE |
|---|---|---|
| 1 | RAM | CS |
| 2 | RAMESH | CS |
| 3 | SUJIT | ECE |

**BRANCH Table**

| BRANCH_CODE | BRANCH_NAME |
|---|---|
| CS | Computer Science |
| EEE | Electrical and Electronics Engineering |

The operation performed using inner join on two tables was STUDENT ⋈ BRANCH and the resulting table is given below.

| ROLL_NO | NAME | BRANCH_CODE | BRANCH_NAME |
|---|---|---|---|
| 1 | RAM | CS | Computer Science |
| 2 | RAMESH | CS | Computer Science |

Apply the operation of left outer join and right outer join on student and branch table. Also provide the output of left outer and right outer join.

13. A hospital database manages patient, doctor, and insurance information using various constraints. Each patient is assigned a unique ID to avoid duplication. The system enforces valid ranges for attributes such as age and restricts deletion of doctor records that are linked to patients. It also applies specific rules to determine insurance eligibility based on predefined conditions.

Illustrate how different types of data integrity are applied in this system and discuss their impact on data accuracy.

14. A college maintains student details in one table and department information with HOD details in another table. The administration needs to combine and analyze data from both tables to manage academic records effectively.

**Student Table**

| StudentID | Name | Dept | Address |
|---|---|---|---|
| 1 | Ravi | CSE | Chennai |
| 2 | Meena | ECE | Coimbatore |
| 3 | Arjun | CSE | Madurai |
| 4 | Kiran | IT | Trichy |
| 5 | Divya | ECE | Salem |
| 6 | Suresh | MECH | Erode |

**Department Table**

| DeptID | DeptName | HODName |
|---|---|---|
| D1 | CSE | Dr. Kumar |
| D2 | IT | Dr. Aravind |
| D3 | MECH | Dr. Suresh |
| D4 | EEE | Dr. Priya |
| D5 | CIVIL | Dr. Ramesh |

Using the given Students **and** Departments tables, determine the output of the following relational algebra operations:

a) Retrieve the HOD name of the MECH department by applying a selection operation on the Departments table.

b) Display only the names of students from the Students table using a projection operation.

c) Using the DeptName values from both tables, determine the resulting tables for the union, intersection, and difference operations.

15. An organization stores important data on a server to ensure continuous access and reliability. To avoid data loss during hardware failures or system crashes, the server maintains duplicate copies of data and performs regular backup operations. These techniques help restore data whenever failures occur.

Outline how replication and backup techniques help in ensuring data protection.

16. A user is working on a presentation using a computer. The application currently in use is stored temporarily for fast processing, while the presentation file is permanently saved on the hard disk for future access.

State the role of primary memory and secondary memory in the above scenario with suitable justification.

| | |
|---|---|
| 17. | A modern computer system uses different memory types to improve processing speed and multitasking performance. Some memory is used for very fast cache operations, while others are used for storing running programs and synchronized data access. Each RAM type is selected based on system performance requirements.<br><br>Justify why SRAM, DRAM, and SDRAM are used for different purposes in a computer system. |
| 18. | A company stores employee documents on a remote server so that staff members can upload files and access them from different locations using network protocols. The system ensures that files are securely transferred and stored on the server.<br><br>Write the steps involved in writing data to server storage in this scenario. |
| 19. | An organization plans to upgrade its storage infrastructure to handle increasing data demands. It needs a solution that provides both high performance and protection against data loss. Different RAID configurations are being evaluated to meet these requirements. The goal is to choose the most suitable RAID level based on speed and reliability.<br><br>Compare RAID 0, RAID 1, RAID 5, and RAID 10 in terms of performance and data reliability. Also, Evaluate the RAID levels and determine which configuration provides the highest data reliability. |
| 20. | An organization uses different servers to support its operations. Employee's access websites through a system that delivers web pages, store and retrieve data from a centralized system, share files within the office network, send and receive emails, and resolve domain names while browsing. Some systems also act as intermediaries to filter content and improve security.<br><br>Classify any five types of servers used in the given scenario and explain the function of each in supporting organizational activities. |
| 21. | A desktop application stores user data in local files. When saving data, the system opens a file, writes the required information, and ensures the file is properly closed. While retrieving data, it opens the file in read mode, reads the contents efficiently, and processes the data for application use.<br><br>a) Illustrate the steps involved in writing data to local storage.<br>b) Describe the process of reading data from local storage.<br>c) Discuss how these mechanisms ensure efficient and reliable data handling. |
| 22. | A company uses cloud storage to access data from different devices, increase storage capacity when needed, and recover important files during system failures.<br><br>Outline any two benefits of cloud storage highlighted in the above scenario with suitable justification. |

| | |
|---|---|
| 23. | A media company needs cloud storage to manage large video files and backups, while another organization requires secure cloud storage with collaboration and compliance features for enterprise use.<br><br>Determine the suitable cloud storage providers for these requirements with justification. |
| 24. | An employee stores important project files in a cloud storage service and accesses them from different locations using the internet. The employee logs into the cloud platform, retrieves the required file, and uses it for further processing. The system ensures secure and reliable access to stored data.<br><br>Write the steps involved in reading data from cloud storage in the above scenario. |
| 25. | Determine the appropriate cloud storage type used in each application with suitable justification.<br><br>A startup company stores its website data and backups using services like AWS and Google Cloud to reduce infrastructure cost and improve scalability.<br><br>A bank maintains its customer transaction data in a dedicated storage environment that is not shared with other organizations due to strict security requirements.<br><br>A multinational company stores sensitive employee records in a private environment while keeping general application data in a public cloud for better scalability. |
| 26. | A cloud service provider offers different storage solutions for various applications. One system allows users to create volumes and partitions like a hard disk, another enables shared file access through a network-attached storage server, and another stores data as objects that are accessed using HTTP requests such as GET and PUT.<br><br>Compare block-based, file-based, and object-based storage systems in terms of their working mechanisms and explain how each type supports different storage requirements. |
| 27. | A company uses cloud storage to manage and access its business data from different locations. Employees upload files through web applications, access stored data through the internet, and rely on the cloud provider to manage backups, redundancy, and storage infrastructure. The organization expects continuous data availability even during hardware failures.<br><br>Explain how cloud storage works in this scenario by discussing the processes of data uploading, storage management, data access, and redundancy/backup mechanisms. |
| 28. | A global company uses a cloud-based storage system to store and manage organizational data across multiple locations. Data is replicated between cloud storage providers to ensure reliability and availability, while migration mechanisms are used to transfer data between providers when required. Users access the cloud services through the internet, and the system connects them to geographically closer cloud servers to reduce response time and improve performance.<br><br>Develop a cloud storage architecture for the above scenario, clearly showing cloud storage providers, users, internet connectivity, data replication, and migration mechanisms. |

29. A database administrator needs to remove all records from the `Employee` table before importing updated employee data for the new financial year. However, the table structure must remain unchanged for future use.

State the DDL command suitable for this operation with justification.

30. Consider the below employee table with the following attributes,

Employee (Eid, Ename, Dept, Email_ID, Phone, Salary)

The below query is used to change the Eid column into Employee_id in the employee table.

Alter table employee rename Eid to Employee_id;

Modify the above query to rename the employee table name into emp.

31. Consider the below employee table to answer the following questions.

| Eid | Ename | Dept | Email_Id | Phone | Salary |
|---|---|---|---|---|---|
| 101 | Adams | Research | adams@gmail.com | 11111 | 50000.00 |
| 102 | Allen | Sales | allen@gmail.com | 22222 | 15000.50 |
| 103 | Blake | Sales | blake@gmail.com | 33333 | 18000.00 |
| 104 | Clark | Accounting | clark@gmail.com | 44444 | 20000.75 |
| 105 | Ford | Research | ford@gmail.com | 55555 | 36000.25 |
| 106 | James | Sales | james@gmail.com | 66666 | 12000.50 |

Predict the schema of the table after executing the below command in employee table.

Alter table employee add age int;

32. A company database contains a table named `Temporary_Employees` that is no longer required for future operations. The database administrator decides to remove the table completely along with all its records and structure from the database.

a) State the DDL command used to perform this operation with suitable reason.
b) Write the SQL query for the operation.

33. Consider the below employee table to answer the following questions.

| Eid | Ename | Dept | Email_Id | Phone | Salary |
|---|---|---|---|---|---|
| 101 | Adams | Research | adams@gmail.com | 11111 | 50000.00 |
| 102 | Allen | Sales | allen@gmail.com | 22222 | 15000.50 |
| 103 | Blake | Sales | blake@gmail.com | 33333 | 18000.00 |
| 104 | Clark | Accounting | clark@gmail.com | 44444 | 20000.75 |
| 105 | Ford | Research | ford@gmail.com | 55555 | 36000.25 |
| 106 | James | Sales | james@gmail.com | 66666 | 12000.50 |

a) Drop the column `phone number` from the Employee table and display the output.

b) Modify the datatype of `salary` column to integer in the employee table and display the output.

34. A college database administrator needs to manage the structure of a student database system. The administrator creates new tables for storing student details, modifies existing table structures to add new attributes, removes unnecessary tables, and renames tables when required.

Illustrate how DDL (Data Definition Language) commands are used in the above scenario and discuss the role of commands such as CREATE, ALTER, DROP, and RENAME in database management.

35. ABC corporate company decides to maintain their employee details in a database. Initially they have stored following details in the employee database with 100 records. The problem with this table is it will allow duplicate values which makes hard to identify each employee uniquely.

Employee (eid, ename, dept, email_id, salary);

a) Write an SQL query to create the above employee table.

b) Modify the above create query to avoid duplicate values as well as NULL values for eid attribute in order to fetch the employee records uniquely.

c) Write an SQL query to add new column named age into the employee table

36. Consider the employee table with the following attributes is created in the scott user. Write the syntax for system user to access the student table from scott user.

Employee(Ename, Dname, Job, Empno, Hiredate, Loc);

37. Consider the employee table with the following attributes,

Employee(Ename, Dname, Job, Empno, Hiredate, Loc)

After updating the employee location to Sweden who is working in sales department.

Is it possible to undo only the department location to Chicago for employees who is working in sales department? Write the syntax for the above requirement.

38. A bank updates customer account balances during a transaction process. To avoid losing all changes during an error, the database administrator uses transaction control mechanisms to partially or completely undo updates when required.

Write SQL queries to illustrate the use of ROLLBACK and ROLL BACK TO SAVEPOINT commands in transaction management.

39. Consider the below employee table to answer the following questions:

**Employee**

| ENAME | DNAME | JOB | EMPNO | HIREDATE | LOC |
|---|---|---|---|---|---|
| ADAMS | RESEARCH | CLERK | 7876 | 23-MAY-87 | DALLAS |
| ALLEN | SALES | SALESMAN | 7499 | 20-FEB-81 | CHICAGO |
| BLAKE | SALES | MANAGER | 7698 | 01-MAY-81 | CHICAGO |
| CLARK | ACCOUNTIN G | MANAGER | 7782 | 09-JUN-81 | NEW YORK |
| FORD | RESEARCH | ANALYST | 7902 | 03-DEC-81 | DALLAS |
| JAMES | SALES | CLERK | 7900 | 03-DEC-81 | CHICAGO |

Write an SQL query to provide privileges to JAMES to perform update operation on employee table. After Completing the task, deny the update permission on employee table for JAMES.

40. Consider the below employee table to answer the following questions:

**Employee**

| ENAME | DNAME | JOB | EMPNO | HIREDATE | LOC |
|---|---|---|---|---|---|
| ADAMS | RESEARCH | CLERK | 7876 | 23-MAY-87 | DALLAS |
| ALLEN | SALES | SALESMAN | 7499 | 20-FEB-81 | CHICAGO |
| BLAKE | SALES | MANAGER | 7698 | 01-MAY-81 | CHICAGO |
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| FORD | RESEARCH | ANALYST | 7902 | 03-DEC-81 | DALLAS |
| JAMES | SALES | CLERK | 7900 | 03-DEC-81 | CHICAGO |

Predict the output for the following SQL queries,

a) INSERT INTO Employee VALUES ('RAM', 'SALES', 'MARKETTING', 7579, '25-MAR-85', 'DALLAS');

b) DELETE FROM Employee WHERE Empno = 7902;

c) UPDATE Employee SET Job = 'FINANCE' WHERE Empno = 7876;

41. A banking system stores confidential customer account details in a database. Managers are given permission to view and update customer information for banking operations. Unauthorized employees should not be allowed to modify sensitive records. The database administrator manages these permissions to maintain data security and controlled access.

Illustrate how GRANT and REVOKE commands can be applied to control user privileges in this system with an example query.

42. Consider the below student table to answer the following questions.

**Student**

| Id | Name | Address | Age | Department | Marks |
|---|---|---|---|---|---|
| 1 | Raju | Chennai | 25 | Arts | 92 |
| 2 | Ramesh | Kashmir | 25 | Engish | 89 |
| 3 | Mani | Delhi | 26 | Arts | 81 |
| 4 | John | Delhi | 26 | English | 85 |

a) Insert two rows into the student table. After entering value, the new table should appear as like the below table by performing SQL Query.

| Id | Name | Address | Age | Department | Marks |
|---|---|---|---|---|---|
| 1 | Raju | Chennai | 25 | Arts | 92 |
| 2 | Ramesh | Kashmir | 25 | Engish | 89 |
| 3 | Mani | Delhi | 26 | Arts | 81 |
| 4 | John | Delhi | 26 | English | 85 |
| 5 | Thiyagarajan | Vizag | 27 | Arts | 91 |
| 6 | Surendran | Pune | 28 | History | 96 |

b) Write an SQL query to perform the following operations,

i) Update the address of student named Mani to Mumbai

ii) Delete the student from the student table whose id is 5.

**Unit 5**

1. A smart home system includes devices such as security cameras, voice assistants, and smart displays. These devices need to communicate efficiently to send alerts, receive commands, and support real-time interaction.

   Analyze the communication requirements of this system and explain which data flow modes are suitable for different devices based on their operation.

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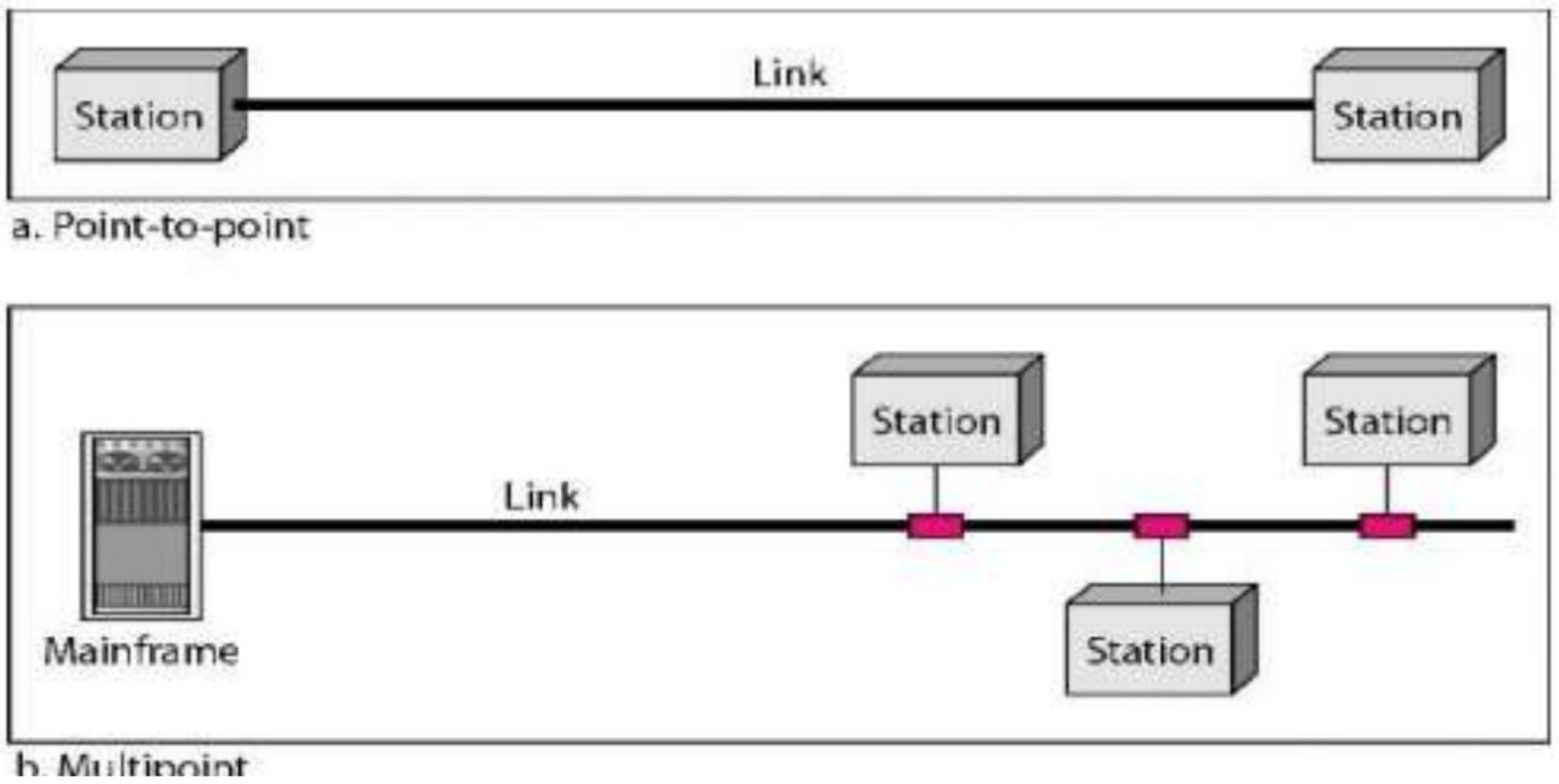
| 2. | A diagram shows two different network connections. In the first diagram, two devices are connected using a dedicated link. In the second diagram, multiple devices are connected through a single shared communication link.<br><br>Analyze both diagrams and explain how data transmission differs in these two types of connections. Also discuss how the communication link is utilized in each case.<br><br>  <br><br><br>Figure: Types of connections:point to point and multipoint |
| 3. | A communication system is designed to support different applications such as data input devices, wireless communication tools, and telephone networks. Each application uses a different mode of data flow to transmit information between devices.<br><br>Analyze how data transmission occurs in different data flow modes and explain how each mode is suitable for the given applications. |
| 4. | A university connects multiple departments using a wired network to enable communication between students and faculty. During file transfer, a student sends a project file intended for a professor, but the file is incorrectly received by another student in a different department.<br><br>Identify the issue in the communication system and indicate the characteristic of data communication that has been violated in this scenario. |
| 5. | A company uses a network to transfer important documents between two offices. During transmission, some parts of the document are altered, resulting in incorrect information at the receiving end.<br><br>Identify the issue in the communication system and indicate the characteristic of data communication that has been violated in this scenario. |

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| 6. | In a data communication system, different types of data flow are used to control the direction of data transmission between devices. Consider a scenario where two devices are connected in such a way that both can transmit and receive data at the same time without waiting for each other.<br><br>Identify the type of data flow used in this system and differentiate it clearly from the other types of data flow used in data communication. |
| 7. | A user sends a video file from a laptop to a smart TV through a network. The data travels through a fiber optic cable, and both devices follow a common set of communication rules to successfully exchange information.<br><br>Identify the components of the data communication system involved in this scenario and explain their roles. |
| 8. | In a network, two devices are connected in such a way that no other device can use their communication link.<br><br>Explain how the communication link is used in this setup. |
| 9. | In a network, all devices are connected to a central device. When a new device is added, the existing network continues to work without interruption. However, when the central device fails, all communication stops.<br><br>Explain why this happens based on the network structure. |
| 10. | A company requires a network where communication between devices should continue even if any one connection fails. The network must ensure uninterrupted data transmission under such conditions.<br><br>Identify the suitable network topology and explain how its structure ensures continuous communication. |
| 11. | A network in an organization is currently implemented using a **bus topology**, where all devices share a single communication line. Due to frequent data collisions and difficulty in troubleshooting, the network administrator decides to convert it into a **star topology**.<br><br>Analyze the changes required to transform the network from bus topology to star topology and explain how the communication process will differ after conversion. |

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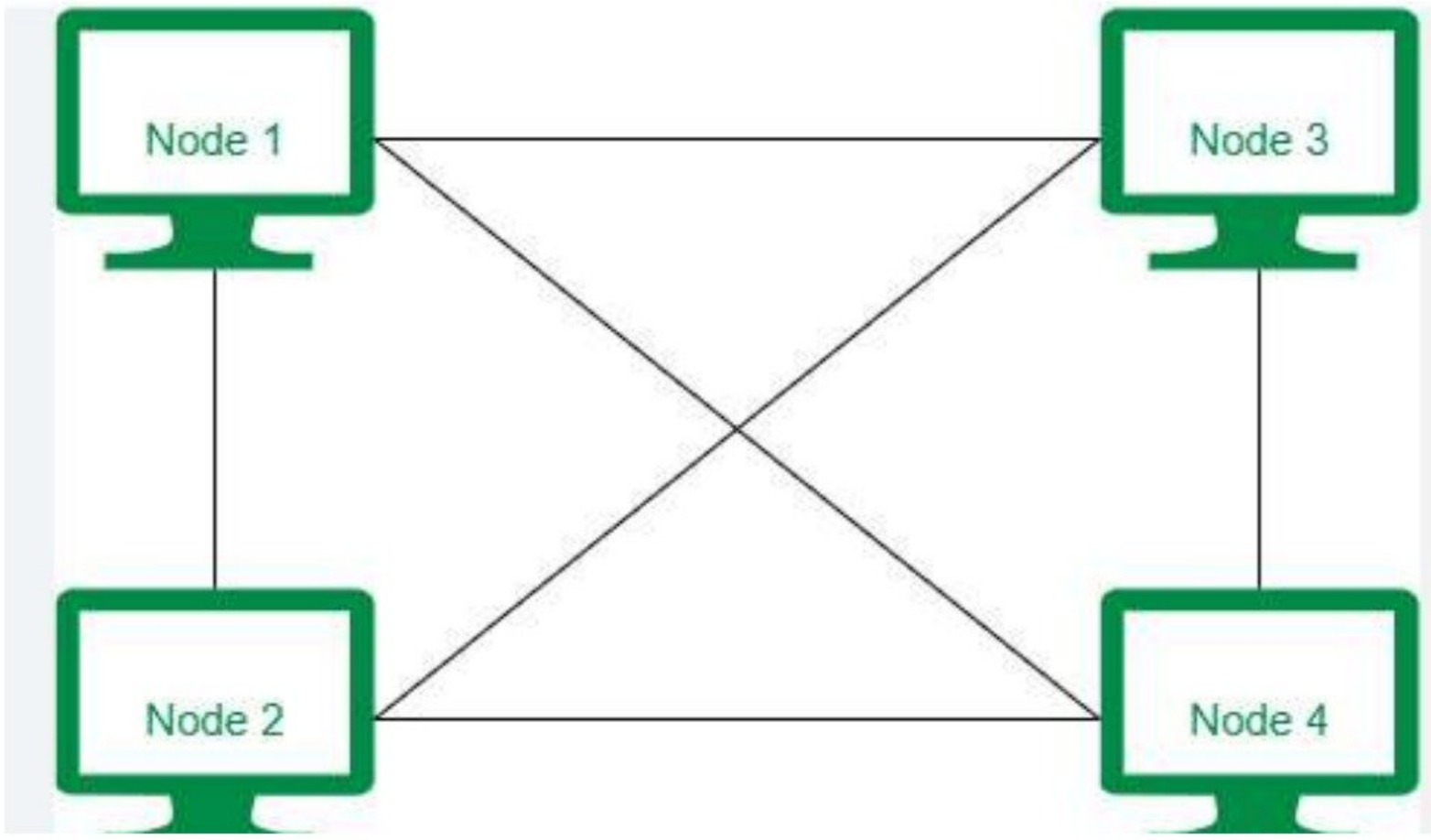
| 12. | A company plans to design a network for its office where communication between devices should be easy to manage, and adding or removing devices should not affect the overall network structure.<br><br>Analyze the requirement and explain which network topology is most suitable. Also describe how this topology supports communication and network management. |
| 13. | A network diagram shows four nodes (Node 1, Node 2, Node 3, Node 4) where each node is directly connected to every other node, forming multiple communication links between devices.<br><br>Analyze how data communication occurs in this network structure and explain how the connectivity between nodes affects reliability and data transmission. Also evaluate how the network behaves if one connection fails.<br><br>  <br><br>Figure:Mesh Topology |

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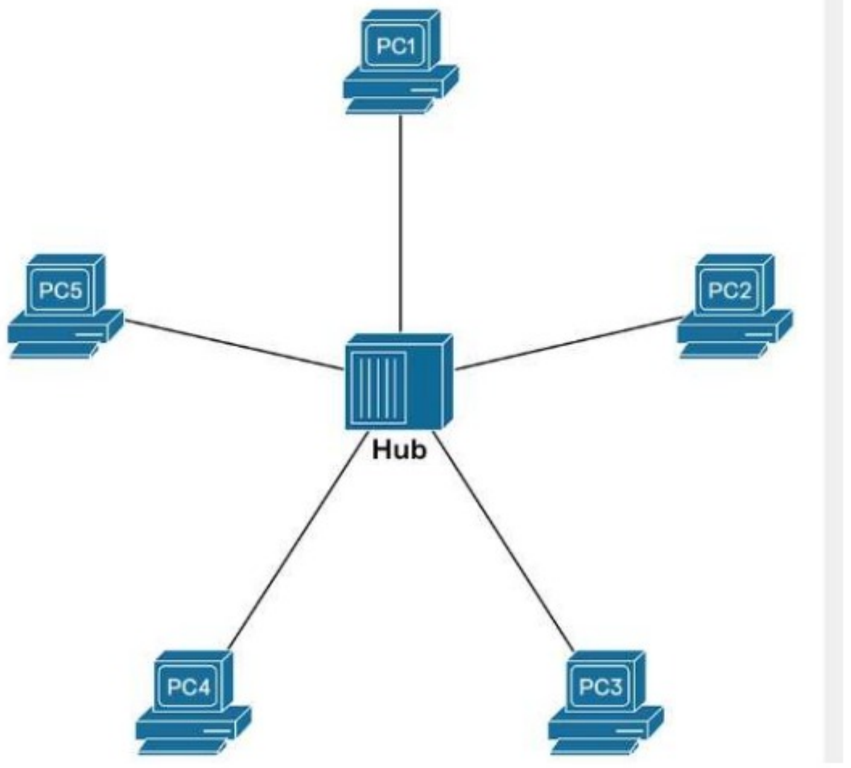
| 14. | 1. A network diagram shows multiple computers (PC1, PC2, PC3, PC4, PC5) connected to a central device (Hub), where all communication between devices passes through this central point.<br>  <br><br>**Figure: Star topology**<br><br>Analyze how data communication occurs in this network and explain the role of the central device in managing communication between nodes. Also evaluate the impact on the network if the central device fails. |
| 15. | A company uses a **Local Area Network (LAN)** within its campus to connect computers, printers, and multimedia devices for internal communication and resource sharing.<br><br>Explain how this LAN helps in connecting devices and sharing resources. |
| 16. | An organization uses a network within its campus to connect computers and other devices for internal communication. In contrast, another network is used to connect devices located in different cities and countries for global communication.<br><br>State one key difference between these two networks in terms of their coverage and usage. |
| 17. | A communication system is designed in two different ways: one where two devices are connected using a direct link, and another where multiple devices communicate through a network with several endpoints.<br><br>Analyze how communication is carried out in these systems and explain how the nature of connectivity influences data transmission. |

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| 18. | A communication system is required to connect devices located in different cities, states, and even countries. The system must support data transmission over long distances and enable communication between geographically dispersed locations.<br><br>Explain why a wide area network is suitable for this requirement. |
| 19. | A network administrator is designing networks for different organizations such as a multinational company, a large organization, and a small business. The administrator must assign suitable IP address classes based on the size and requirement of each network.<br><br>Analyze how IPv4 address classification supports this requirement and explain how different classes are suitable for networks of varying sizes and purposes. |
| 20. | A company is assigned the IP address **192.168.10.25** with subnet mask **255.255.255.0**. Identify the network address and host address, and indicate how the given IP address is divided into network and host portions using the subnet mask. |
| 21. | A network administrator is configuring an internal communication system for a large organization. The system uses the IPv4 address **10.0.5.20** with a subnet mask of **255.255.252.0** to manage multiple departments efficiently. The administrator needs to analyze the addressing structure to ensure proper routing and allocation of hosts within the network.<br><br>Based on this configuration, identify the class of the given IP address, determine the network address and broadcast address, calculate the total number of possible hosts in the network, and indicate whether subnetting is applied in this case |
| 22. | A company assigns subnets A, B, C, and D from the network 192.168.12.0. Each subnet must support communication within its range without overlap.<br><br>Identify the network addresses of all four subnets. |
| 23. | In a small office network, the administrator manually assigns IP addresses to printers and servers, while user devices automatically receive IP addresses from a central system.<br><br>Identify which devices use static IP assignment and which devices use dynamic IP assignment. |
| 24. | In an organization, network devices that require постоян connectivity are configured manually, while temporary devices are assigned IP addresses by a network service.<br><br>Indicate the type of IP assignment used for permanent devices and temporary devices. |

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| 25. | A network administrator divides the network 192.168.12.0 into four equal subnets (A, B, C, and D) to allocate IP addresses efficiently across different departments. Each subnet has a defined network address and broadcast address.<br>Based on the given subnetting, identify the network address of subnet B, the broadcast address of subnet C, and determine the subnet in which the IP address 192.168.12.200 belongs. |
| 26. | A company is setting up a new network where multiple departments are connected through routers. The DHCP server is located in a different network segment, so a DHCP relay agent is used to forward requests from client devices. The network administrator must configure the system to ensure automatic IP address allocation to all clients.<br>Based on this setup, outline the steps required to configure DHCP, including creating a DHCP pool, defining the network and subnet mask, and specifying necessary network parameters. Also, indicate the role of the DHCP relay agent in this scenario. |
| 27. | A company network initially fails to assign IP addresses to remote clients located in another subnet. After analysis, it is found that the DHCP server is not directly reachable.<br>Determine the configuration requirements needed to resolve this issue and justify the role of each step in DHCP setup. |
| 28. | In a large enterprise network, improper DHCP configuration leads to issues such as clients not receiving IP addresses, inability to access external networks, and failure in domain name resolution.<br>Describe the complete DHCP configuration process required to resolve these issues and justify the importance of each parameter used. |
| 29. | In a campus network, the **access layer** connects multiple computers in a lab through switches for daily student usage.<br>**Explain the role of the access layer in supporting end-user devices.** |
| 30. | A campus network uses **load balancing** to send data through multiple paths instead of a single path to avoid congestion.<br>**Explain how load balancing improves network performance.** |
| 31. | A user enters the website address www.campus.edu in a browser, but the system needs to find the corresponding IP address to establish communication. The request is processed through different levels of DNS servers before obtaining the final result.<br>Identify the type of DNS translation used and outline the role of each DNS server involved in resolving the domain name. |

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| 32. | A small office network uses a **private IP address** like 192.168.1.10 for internal communication, while the router is assigned a **public IP address** by the Internet Service Provider to access the internet.<br><br>**Classify the given IP addresses and distinguish their roles in terms of accessibility and usage.** |
| 33. | A student attempts to access the website www.campus.edu from a campus network. The system must resolve the domain name into an IP address using the DNS hierarchy. During this process, multiple DNS servers are involved before the final IP address is obtained.<br>Explain the step-by-step process of domain name resolution, clearly describing the role of root servers, TLD servers, and authoritative servers. Also, distinguish between forward and reverse domain name services. |
| 34. | A computer **A1 in LAN A** needs to send data to **computer D1 in LAN D** in a campus network that follows a three-layer architecture (access, distribution, and core).<br><br>Explain the step-by-step path followed by the data from source to destination. Clearly indicate the role of the access layer, distribution layer, and core layer at each stage of transmission. |
| 35. | A university is designing a campus network that connects multiple buildings such as academic blocks, administration offices, and hostels. The network is structured using core, distribution, and access layers to ensure efficient communication and scalability. During operation, issues such as data congestion, uneven traffic distribution, and slow communication between departments are observed.<br><br>Explain how the core, distribution, and access layers function in this campus network. Also, describe how data flows between different LANs and discuss the role of load balancing in improving network performance. |
| 36. | A network administrator plans to upgrade a system from IPv4 to IPv6 to support a large number of devices and improve addressing capability.<br><br>**Justify how IPv6 provides advantages over IPv4 in this scenario.** |
| 37. | A campus network core handles a large volume of traffic and must ensure continuous high-speed data transfer between departments.<br><br>**Assess the impact of an inefficient core layer on overall network performance.** |
| 38. | A campus network is divided into three layers to improve performance and manage traffic efficiently. Data from one department is transferred to another through intermediate devices.<br><br>**Explain how data flows across different layers in a campus network.** |

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| 39. | An organization allows employees to access company applications using personal mobile devices. It also restricts access to harmful websites, divides the network into segments, and ensures that only authorized users can access internal resources.<br><br>**Identify any three types of network security used in this scenario and state their purpose.** |
| 40. | **A university is designing a campus network to connect multiple departments located in different buildings. The network must support high-speed data transfer, apply routing policies between departments, and provide reliable access to end-user devices like computers and printers.**<br><br>**Analyze how the core layer, distribution layer, and access layer contribute to the efficient configuration and management of the campus network.** |
| 41. | **A network administrator is analyzing different IP addresses assigned to various systems in an organization and across the internet. The administrator needs to identify the class of each IP address to understand their usage and network design.**<br><br>**The given IP addresses are:**<br><br>**1. 10.25.14.8**<br>**2. 172.20.5.6**<br>**3. 192.168.1.1**<br>**4. 224.0.0.5**<br>**5. 240.0.0.1**<br><br>**Classify each IP address into Class A, B, C, D, or E and justify your classification based on the first octet range.** |

42.

1. A multinational company operates offices in different cities and countries. The head office needs to communicate with two remote branches using a dedicated communication link for secure data transfer. At the same time, multiple other branch offices are interconnected through a shared network infrastructure that allows data exchange between many locations.

Based on this scenario,
**identify and differentiate the types of WAN connections used** for:

1. Communication between the head office and two specific branches
2. Communication among multiple branch offices

Also, **justify which WAN type is more suitable for global communication backbone and why.**

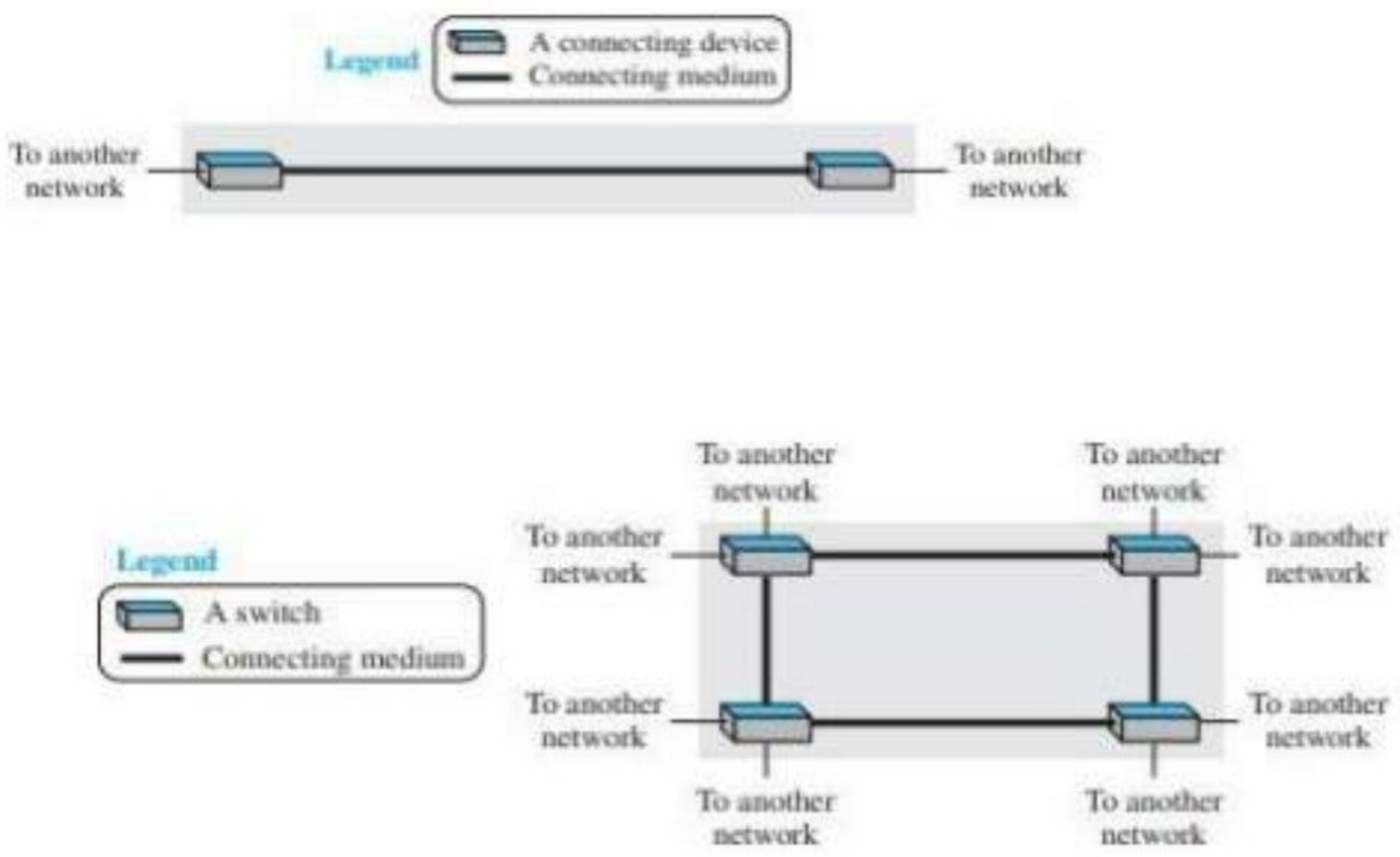


**Figure : Types of WAN**